

Endgames on an $n \times n$ board

the depth laws and where they fail, the positions that take longest, a rule that mates without a table, two endgames whose deepest win runs away until the win itself disappears, and what happens when mate is replaced by the capture it stands for

Summary of findings

Every figure is computed from a complete table: an exact distance for every legal position, both sides to move. Two rule sets run throughout, ordinary chess and the *capture rule* of §10; diagrams are badged N or C accordingly, and where there is a line to follow the controls under the board step a ply at a time.

Depth obeys a law, except when it does not. Deepest KQK is mate in $\lceil 3(n-2)/2 \rceil$ for every $9 \leq n \leq 60$, $n=8$ the sole exception. KRK has no such expression, but a recurrence of period six, and it fails once, at $n=24$. KBNK is quadratic and zig-zags with the parity of the board, an odd board's corners being all one colour while the mate needs one the bishop can reach. Square colour settles KBBK outright: the drawn positions are exactly the same-coloured ones. (§1.1, §5.2.3)

A queen needs no table. Six rules, a pure function of the position with no lookup and no dependence on n , mate from every legal position on every board from 4×4 to 14×14 — 13 559 524 of them — at worst 1.79 times the optimum. (§2.2.3)

Arm the defender and the win can vanish. KQKR grows 35, 44, 54, 69, 85, 108, 132 moves for $n=8 \dots 14$, then **218** at $n=15$; at $n=16$ a whole class stops being winnable, the drawn fraction jumping from 0.13% to 26.6% while the deepest win *falls*. (§7)

Two black kings are easier to mate than one. Under the strict mating rule White wins every legal position at every size from 3 to 17, and the 8×8 mate is in **7** against 10 for the same queen against one king: the method is a fork, not a corner, and in 56.8% of the mates the white king contributes nothing. (§8)

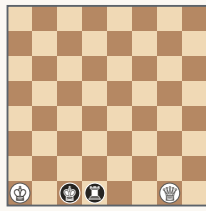
Play for the kings instead and it becomes a fight. Under the capture rule the same four men give White 65.9% of the placements the mating rule gives him all of, Black holding a third and winning a few. Two computed facts carry it: **two kings hunt down a lone king** from every placement on every board up to 14×14 , and $K+Q$ vs K is won everywhere and a move faster than ordinary KQK. The first has a **critical board size** — at 15×15 the hunt fails, 93% of its wins turn drawn, and the deepest win drops from 62 moves to 37 and stays there to $n=22$. And the rules are conservative rather than new: with one king a side they are chess with a single clause replaced, stalemate becoming a loss except in totally immobile positions, where a *checkmate* becomes a draw instead. (§10)

Take the queen away too, and three kings beat two. White wins from every white-to-move placement on every board from 3×3 to 11×11 , 1 987 925 940 of them at $n=11$. The endgame turns on one row of the census: three kings against one is won with *either* side to move, so any capture White can make wins on the spot. It is the only table here in which material falls on both sides, it contains no zugzwang, and the clause it rests on expires at 15×15 . (§9)

“Two knights cannot mate” is false as stated, and it costs a table. $K+N+N$ vs K holds **120** mates on 8×8 , $16n-8$ on every board; what is true is that Black is never in *zugzwang*, the deepest white win being one ply at every size from 3 to 16. So three knights need a KNNK table underneath them. Under the capture rule — for this material exactly chess with stalemate scored as a loss — the same two knights go from 616 won placements on 8×8 to **5 713 976**, then collapse in their turn: 51, 61, 75, 91, 111, 151, **219** plies for $n=8 \dots 14$, and 0.45% at 15×15 . (§6)

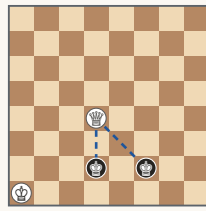
What the cross-checks were for. Three of the bugs they caught were plausible numbers rather than crashes. A successor counter is *unsound* on a symmetry-reduced index. Treating “the king steps onto a piece” as a capture is sound with one white piece and wrong with two: it cost KBNK two moves on 8×8 and hid in KBBK for as long as KBBK existed. And a table that *converts* cannot be swept until nothing changes. (§11)

Five positions worth keeping.



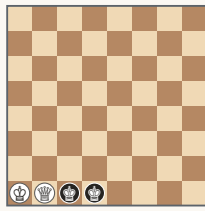
○ [N]

KQKR
mate in 26 at $n = 8$, in
176 at $n = 15$, **drawn** at
 $n = 16$



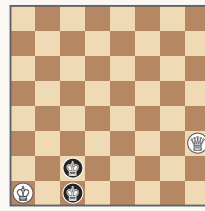
○ [N]

KQKK, mating rule
mate, and the white king
is irrelevant to it



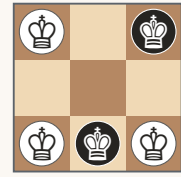
● [N]

KQKK, mating rule
no move, one king in
check: mate by the
ordinary rule, a **draw**



○ [C]

KQKK, capture rule
nothing hanging, White
to move, and **lost** in 4
plies



○ [C]

KKK vs KK
three kings against two,
and **whoever moves**
first wins

A fortress whose depth runs away until the win disappears; a queen mating two kings by herself, on two rays at once; what “both at once” costs, a position this rule declines to call mate; a queen on the board, nothing hanging, and White lost because two kings that guard each other have sealed his king in the corner; and five kings on nine squares, whoever moves first winning.

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1 The endgames, and what the tables hold

One section per holding, each asking the same two questions: what does ordinary chess make of it, and what does §10's capture rule? Every table is complete. The first four leave Black a lone king and White one or two long-range men, and every design decision here was made for them; the knights of §6 leave Black just as bare but break those decisions from *White's* side; the last three do not leave Black bare at all — §7 gives him a rook, §8 a second *king*, and §9 takes the pieces away entirely, leaving five kings and the capture rule as the only game there is.

	White has	8×8 deepest	largest n	entries/side there
■ KQK	king and queen	mate in 10	60	5 820 822 000
■ KRK	king and rook	mate in 16	54	3 092 041 836
■ KBBK	king and two bishops	mate in 19	22	3 375 083 250
■ KBNK	king, bishop and knight	mate in 33	19	2 093 606 865
■ KQKR	king and queen <i>against king and rook</i>	mate in 35	17	856 925 460
■ KQKK	king and queen <i>against two kings</i>	mate in 7	24	6 667 930 368
■ KNNK	king and two knights	mate in 1	16	260 467 200
■ KNNNK	king and three knights	mate in 21	15	11 591 361 600
■ KKKvKK	three kings <i>against two</i>	63 plies [†]	11	1 987 925 940 [†]

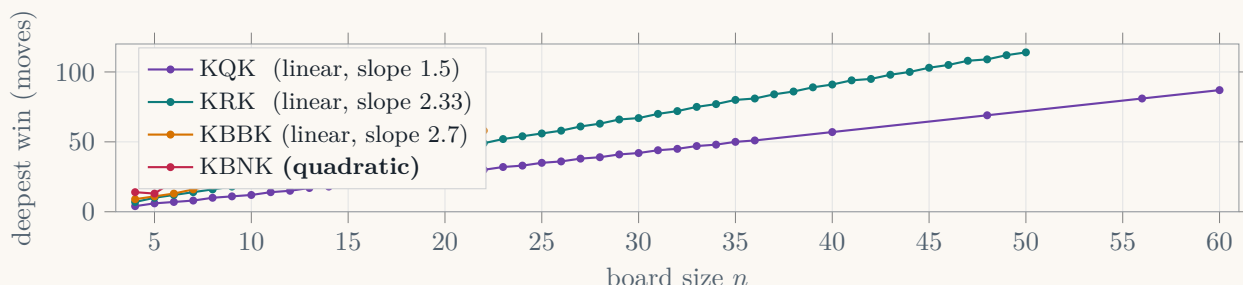
[†] The last holding has no mate to be deep in: its depth is plies to the capture of the second black king, its limit is the largest board built rather than the largest that fits, and its last column counts *placements* rather than the symmetry-reduced entries above.

The invariant that makes them tractable

Every capture Black can make leaves material that *cannot mate*, so it is an immediate and permanent draw: $\dots K \times Q$ and $\dots K \times R$ leave bare kings, $\dots K \times B$ in KBBK leaves KBK, and in KBNK $\dots K \times B$ leaves KNK while $\dots K \times N$ leaves KBK. Consequently no position has a predecessor outside its own table: the unmove generator needs **no un-captures at all**. KNK and KNNK join them, and KNNK is the sharper case, since $\dots K \times N$ leaves KNK, in which **no mate exists on any board size**.

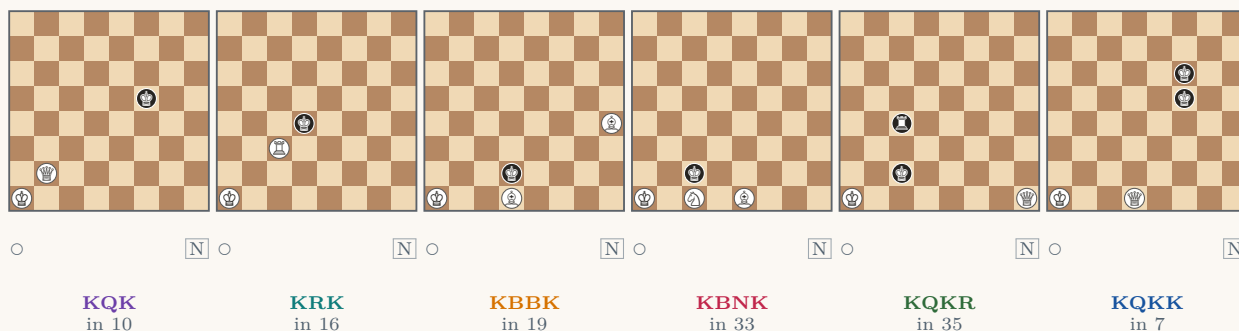
This is exactly where it stops, and knights mark the boundary more finely than rooks. KRRK looks like KBBK and is not: taking one rook leaves KRK, which White wins. KNNNK looks like KBBK too and is also not, but only just — taking one knight leaves KNNK, which White wins from 1 448 of 1.9 billion positions at 16×16 . Rare is not never, and §6 is what that costs. §8 then shows “bare” is the wrong word, a second black *king* satisfying every clause above. What breaks the apparatus is Black being able to *do* something.

1.1 Every depth in this document, on one axis



Three of the four are straight lines. KBNK is not, and it zig-zags because even and odd boards behave differently. Both facts have one cause (§5.2.3).

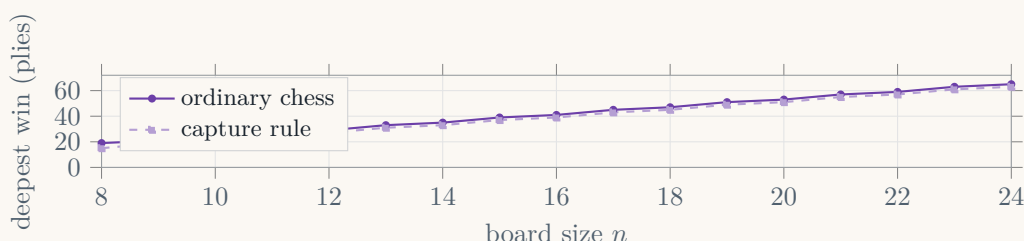
The hardest 8×8 position in each endgame. Same board, six different answers.



2 King and queen

The simplest endgame with a forced mate, and the only one here that can be played correctly with no table at all.

2.1 Growth



One line shifted down by exactly two plies from 9×9 up: 2.96 plies a board for chess against 3.00 for the capture rule, residuals 0.57 and 0.50.

2.2 Normal chess

2.2.1 Deepest mate

KQK depth law

The deepest KQK win on an $n \times n$ board is

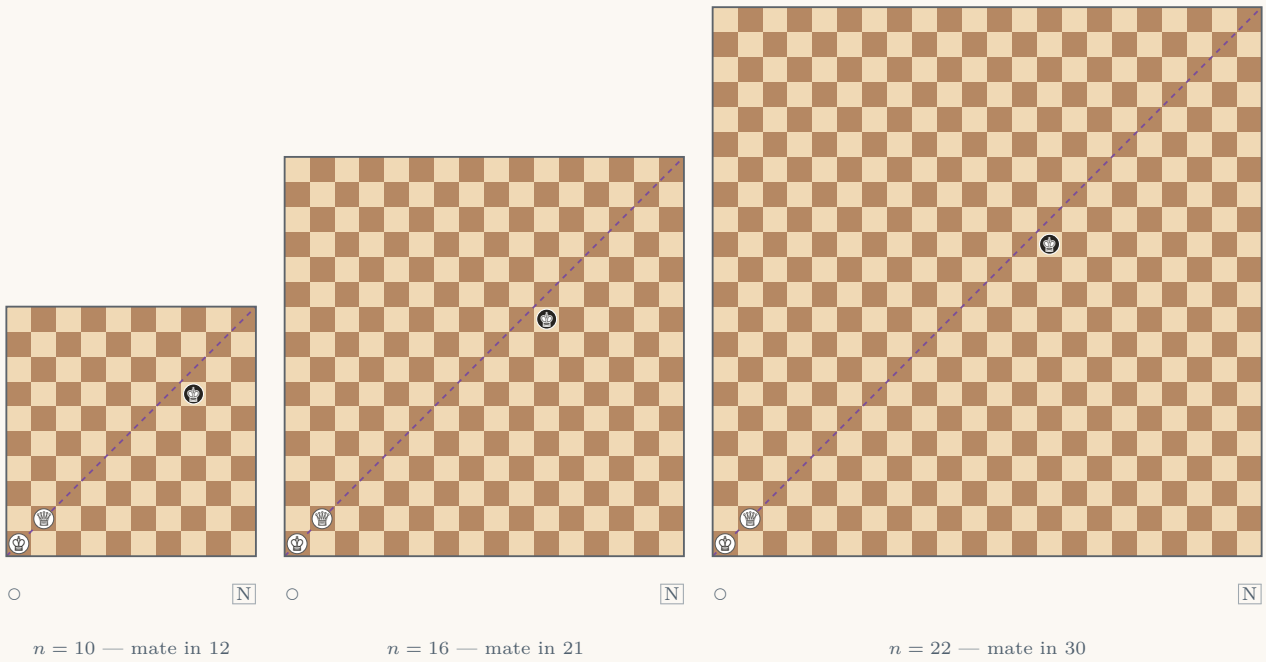
$$\text{mate in } \left\lceil \frac{3(n-2)}{2} \right\rceil$$

for every n from 9 to 60 computed — three plies per unit of board size, exactly what the shrinking-box technique predicts.

The ordinary chessboard is the sole exception. At $n = 8$ the formula gives 9; the true value is 10, attained by a *single* position up to symmetry.

2.2.2 Examples

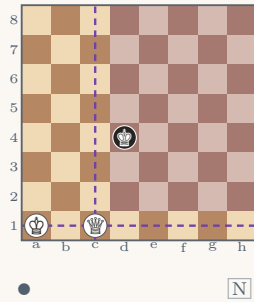
White king in the corner, queen beside it, black king out past the middle: wK $a1$, wQ $b2$, bK $(\lfloor n/2 \rfloor + 2, \lfloor n/2 \rfloor + 1)$, verified deepest for every n from 9 to 32. The black king's square advances one diagonal step every two board sizes, the same $\lfloor n/2 \rfloor$ that makes the law alternate between $3n - 7$ and $3n - 6$.



Dashed, the long diagonal; drawn at a common square size, so what grows across the three is the board and not the position. At $n = 8$, the law's exception, the black king sits one square nearer: $wK a1$, $wQ b2$, $bK f5$, unique up to symmetry.

2.2.3 Mating without the table

At $n = 60$ the KQK table holds 5 820 822 000 entries a side. Is there a *fixed rule* — no dependence on n , no memory, no lookup — that mates from everywhere and is not much slower? There is.



the queen's file and rank, and the **cage** they cut out

If the queen shares neither file nor rank with the black king, her lines cut the board into four rectangles and he sits in one. To leave it he would have to step onto a line she attacks every square of, so **Black cannot enlarge the cage**: only White can, and only by moving the queen. That asymmetry is the whole algorithm.

With R the cage and T the *post* the white king walks to, $\Phi = (|R|, \text{cheb}(wK, T), \text{man}(wK, T))$ ordered lexicographically is unchanged by every black move, so a policy that only lowers Φ cannot cycle. One caveat, checked rather than assumed: the white king can stand on the queen's line and block it, so every candidate move is tested against all of Black's replies.

Call a White move **sound** when, after it, Black cannot capture the queen, Black has a legal move, the black king is on neither the queen's file nor her rank, and no black reply either changes the cage or leaves Black with no move. White plays the first rule that applies.

-
- 1 **Mate**, if a move mates.
 - 2 **Finish**, once width plus height is at most 6: for $k = 1, \dots, 5$ in turn, if White has a forced mate in at most k using only sound moves that never grow the cage, play its first move.
 - 3 **Squeeze**: a sound move that strictly shrinks the cage, smallest area first and at equal area the squarest.
 - 4 **Walk**: a sound move that keeps the cage and brings the white king strictly nearer the post.
 - 5 **Wait**: a sound move that keeps the cage, hemming the black king in most.
 - 6 Any legal move that neither hangs the queen nor stalemates.
-

Rules 3 and 4 lower Φ , which is bounded below, so neither can repeat a position. The *iterative deepening* in rule 2 is not decoration: a fixed depth lets “mate in at most five” stay true forever while the pieces

shuffle. Rules 5 and 6 carry no such argument, which is why the rule is scored by machine; over all 144 508 white-to-move 8×8 positions rule 5 fires 96 times and rule 6 464, against 132 928 uses of rule 3.

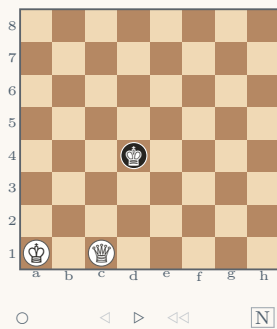
One repair, and it took a 13×13 board to find. With the corner post alone the rule hangs from 53 661 positions there: the cage is a single file eight squares long, so White is winning by every measure, and his king has reached a post two steps in from the corner where it is useless, the black king being at the *other* end of the strip. The cure is to send the king where the work is — once the strip is seven or more long the post moves beside the square the queen wants. Both posts are functions of the cage alone, so Φ survives. All 4 466 964 positions up to 12×12 pass with the corner post and say nothing about the defect.

White’s move is fixed to the rule, Black plays anything, and the worst case comes from backward induction over the induced graph. The ratio to the table’s optimum runs 1.75, 1.57, 1.60, 1.75, **1.79**, 1.73, 1.76, 1.72 over $n = 4, 6, 8, 10, 11, 12, 13, 14$, with no failure at any size.

A rule that mates from everywhere, at most $1.79\times$ the optimum

For every board from 4×4 to 14×14 the rule mates from every legal position — 13 559 524 in all — and never takes more than $1.79\times$ as long as the table. Its worst case grows like $5n/2$ against an optimum near $3n/2$, so throwing away five billion entries costs a factor tending to $5/3$.

The control. Substituting the tablebase’s own optimal move for the rule, in the same harness, reproduces the optimum exactly: ratio 1.00, zero failures. That is what says the scoring measures the policy and not the scorer.



wK a1, wQ c1, bK d4 — mate in 10 with the table, 16 with the rule

The hardest 8×8 position for the rule. Both phases are plain in the cage size: ten queen moves take it from 35 squares to 3, five king moves walk the white king in, and the eleventh queen move mates. Note move 1, past the black king to g5 rather than towards it, and that the queen stands still from move 11 to 15.

Two clauses that look unnecessary, and are not. *Never put the black king on the queen’s file or rank:* a check along a line dissolves the cage, the king being momentarily on both sides of it, while a diagonal check does not and is often the fastest squeeze there is. *Never let Black reach a square with no move:* every legal White move must then hand him a flight square, which usually means opening the cage. Without it the rule loses its way from 49 positions on 8×8 ; with it, from none.

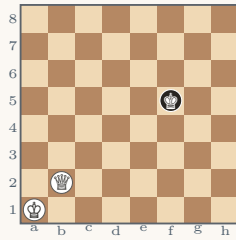
2.3 Capture the king

2.3.1 Deepest win

A convention, once, for every capture table here. These tables end when the mover has no chess move — mate or stalemate, and §10.1 makes both a loss. The solvers of §8 and §9 play on until the king is physically taken, two plies later, so a depth quoted there is two more than the same depth quoted here.

The capture rule is worth exactly one move, and not a move more. §2.2.1’s closed form carries straight over: chess takes $2\lceil 3(n-2)/2 \rceil - 1$ plies and the capture rule $2\lceil 3(n-2)/2 \rceil - 3$. The one move is the stalemate dodge — ordinary KQK’s longest win spends a tempo avoiding a stalemate it is not allowed to want, and the rule refunds that tempo, once, whatever the board size.

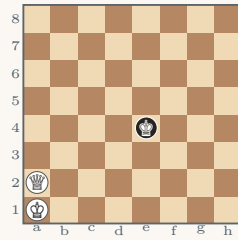
2.3.2 Examples



○

[N]

chess: Ka1 Qb2 vs kf5
19 plies



○

[C]

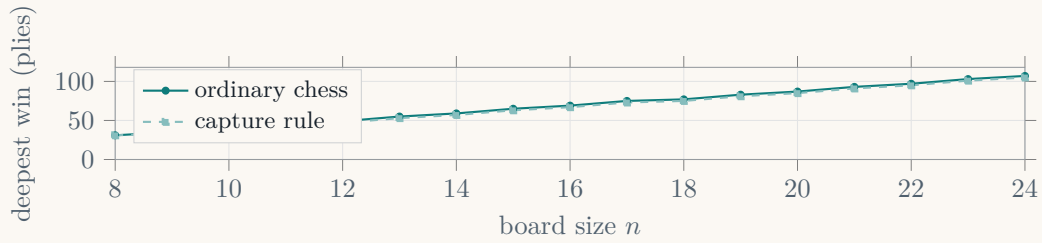
capture: Ka1 Qa2 vs ke4
15 plies

The deepest position changes as well as the depth. Under the capture rule the queen starts *beside her own king* rather than on the long diagonal: she no longer needs room to give the black king a flight square, there being no reason to give him one.

3 King and rook

The same solver and the same index, with one line of the move generator changed: a rook is a queen restricted to the four orthogonal rays.

3.1 Growth



Two plies apart from 9×9 up, at 4.76 plies a board against 4.72. But **at $n = 6, 7$ and 8 the capture rule saves nothing at all:** 8×8 KRK is 31 plies under both games, the opposite of the queen's answer. A rook's mate is a squeeze along a rank with the kings in opposition, and on a board that small the squeeze never passes through a position where White would have preferred a stalemate.

3.2 Normal chess

3.2.1 Deepest mate

KRK depth law

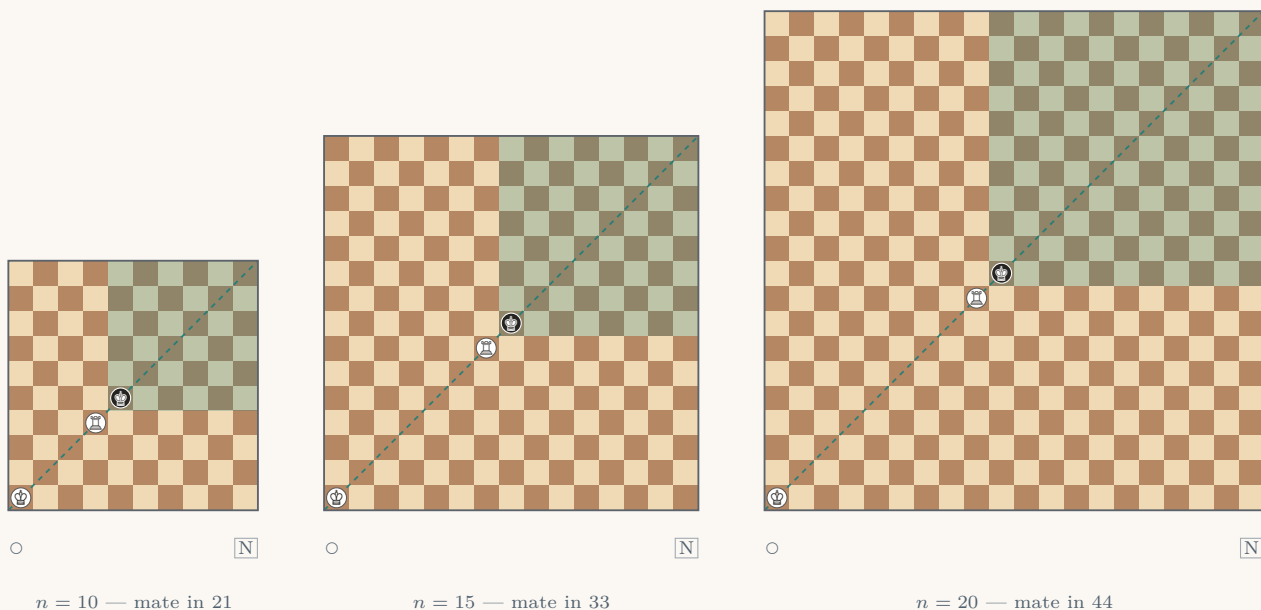
No expression $\lceil (an + b)/c \rceil$ fits, searched exhaustively for $a < 60$, $|b| \leq 60$, $c \leq 24$. What holds instead is a recurrence of period six:

$$m(n) = m(n - 6) + 14, \quad \text{i.e.} \quad m(n) = 14 \left\lfloor \frac{n}{6} \right\rfloor + c(n \bmod 6), \quad c = (-3, 0, 2, 5, 7, 10)$$

for every $12 \leq n \leq 50$ computed, with **exactly one** disagreement, at $n = 24$, where the law predicts 53 and the true value is 54.

3.2.2 Examples

All three men on the long diagonal, the black king on the *centre square*, the rook between it and the corner: wK $a1$, rook (k, k) , bK $(k+1, k+1)$, $k = \lceil n/2 \rceil - 2$. Verified deepest for every n from 8 to 32, $n = 24$ included, where the depth law itself fails.



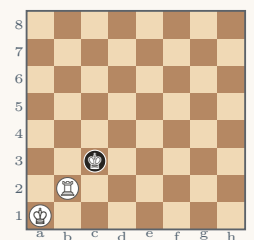
The tinted quadrant is what the rook's file and rank already confine Black to. The rook stands next to the *black* king, the opposite of KQK: a queen confines by radiating from anywhere, a rook by *being* the fence, so the hardest position is the one where the fence already stands at its widest useful setting and the white king still has the whole diagonal to walk.

3.3 Capture the king

3.3.1 Deepest win

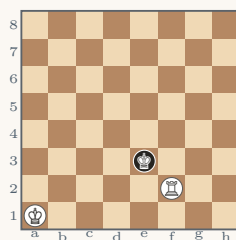
Equal depth does not mean the same position. Both maxima below are 31 plies on 8×8 , and only the first is the one the textbooks describe.

3.3.2 Examples



○ N

chess: $Ka1 Rb2$ vs $kc3$
all three men in the corner



○ C

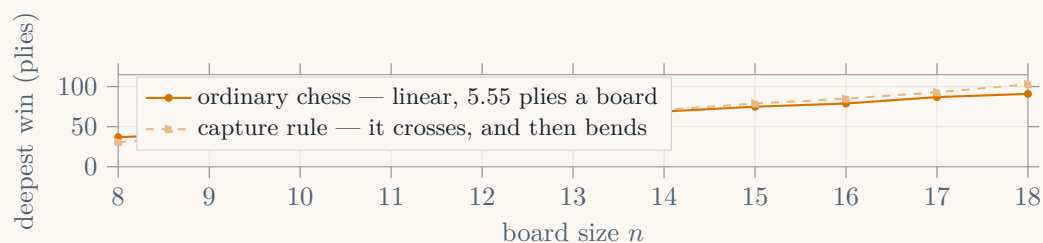
capture: $Ka1 Rf2$ vs $ke3$
the rook thrown across

The chess maximum is §3.2.2's placement at $n = 8$; the capture maximum puts the rook five files away. Two different 31-ply problems out of the same material.

4 King and two bishops

The first endgame here with two white men, and the first in which the *colour* of a square decides the result outright.

4.1 Growth



The capture rule helps White, then stops helping and starts costing

Two bishops are the first material here where the difference *changes sign*: $-6, -4, -4, -2, 0, 0, +2, +4, +6, +6, +12$ plies for $n = 8 \dots 18$.

A rule that only turns draws into wins cannot make an existing win slower. What it can do is add *new* wins, and a new win can be deeper than every old one: positions where White can force stalemate but not mate are chess draws, they become capture wins, and on a big enough board the hardest of them is harder than anything he could already mate. The maximum stops describing the same task, and the fit shows it — chess is linear at 5.55 plies a board, the capture rule needs a quadratic.

4.2 Normal chess

4.2.1 Deepest mate

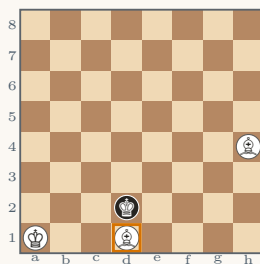
Mate in 19 on the ordinary board, the published figure, not fitted to. This is the only white material here with no closed form, no recurrence and no parity wobble, so the interesting part is which positions attain it.

An exhaustive search that finds nothing

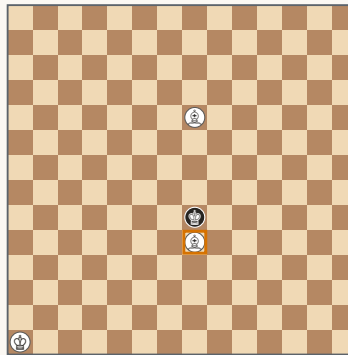
Allow each of the six coordinates to be a constant measured either from the $a1$ corner or from the far edge (c or $n - 1 - c$), and seed from every deepest position with both bishop orderings. Then over $n = 9 \dots 14$ the surviving templates run $42\,624 \rightarrow 464 \rightarrow 240 \rightarrow 68 \rightarrow 2 \rightarrow 0$; over $n = 12 \dots 18$ the family dies at $n = 17$; even boards only, it dies at $n = 18$; odd boards only, at $n = 17$. KQK and KRK each have a template holding for 25 consecutive board sizes. KBBK has none.

4.2.2 Examples

What *is* invariant is a motif: the white king in a corner and one bishop beside the black king **undefended**, so White is a tempo down and must rescue it first. That holds for all 15 classes at $n = 8$ and all 7 at $n = 14$, and at every n an adjacent bishop in a deepest position is never defended.



○ N
 $n = 8$ — mate in 19,
 one of only 15 classes



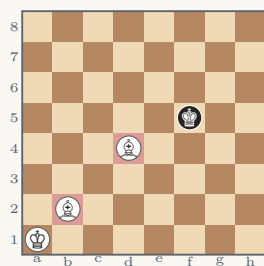
○ N
 $n = 14$ — mate in 35,
 one of only 7 classes

The ringed bishop is the hanging one. Note how little the two boards have in common beyond that.

4.2.3 Where square colour decides the game

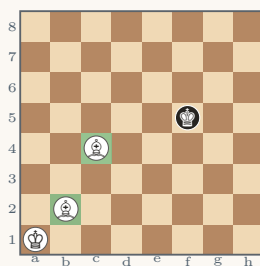
Computed, not assumed

Every legal white-to-move position with both bishops on *one* colour is drawn, for every n from 3 to 18 without exception. On 8×8 that is 2 577 900 of 5 082 028 legal white-to-move placements, and the whole of the white-to-move draw count. Same-coloured configurations are indexed, initialised and swept like any other; they simply come out drawn. Splitting the table by bishop colour would have halved it, and *assumed* the result.



○ [N]

both bishops dark —
drawn, always



○ [N]

opposite colours —
a win

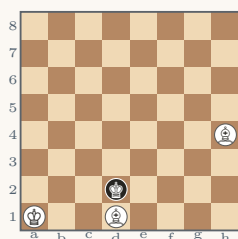
The converse very nearly holds too. With opposite colours White wins except for **520** placements on 8×8 , 65 up to symmetry, every one with a bishop trapped beside a cornered king: every rescue stalemates, everything else drops the bishop.

4.3 Capture the king

4.3.1 Deepest win

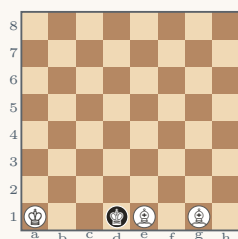
At 8×8 , where the capture rule is still saving time, the deepest win is 31 plies against 37. Its two bishops stand on the *same* colour, and so do the deepest at every size: what the rule repeals is §4.2.3 itself (§4.3.3).

4.3.2 Examples



○ [N]

chess: Ka1 Bd1 Bh4 vs kd2
37 plies



○ [C]

capture: Ka1 Be1 Bg1 vs kd1
31 plies

Under the capture rule both bishops come home to the back rank beside their own king; under chess one is thrown to *h4* to cover the corner the mate needs.

4.3.3 Both bishops on one colour

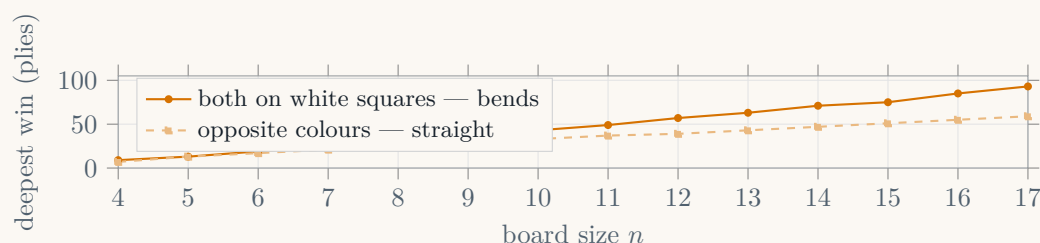
The capture rule repeals the colour rule outright

Every position §4.2.3 finds drawn is a capture-rule **win**. With both bishops on white squares White wins *every* legal white-to-move placement on every board from 4×4 to 17×17 — 715 394 384 of them at $n = 17$ — and the same-coloured positions are the *deeper* half, 31 plies against 25 at 8×8 and 93 against 59 at $n = 17$. Only 3×3 resists, in 4 placements of 96.

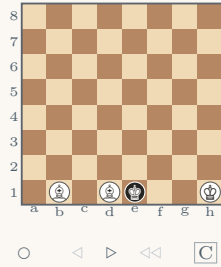
§4.2.3 survives as stated: the bishops still never mate. **Every one of these wins is a stalemate**, and two laws hold without exception on every board computed — the black king stands **on the border**, and **on the colour the bishops do not have**.

n	4	5	6	7	8	9	10	11	12	13	14	15	16	17
deepest (plies)	9	13	19	25	31	35	43	49	57	63	71	75	85	93
stalemate wins	194	720	894	2448	2442	5792	5174	11280	9426	19440	15534	30800	23834	45888

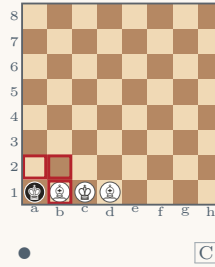
The stalemate count is an exact cubic in each parity, $11n^3 - 29n^2 + 13n + 5$ for odd n and $7n^3 - 20n^2 + 18n - 6$ for even, fitted to $n \leq 17$ and confirmed out of sample at $n = 18$ and 19.



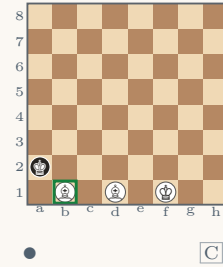
The two curves cross at $n = 5$ and never meet again. Opposite colours are a straight line, 3.85 plies a board with residuals 0.78, which a quadratic does not improve. White squares climb at 6.47 and bend: residuals 1.80 against a line, **0.89** against a quadratic, with an n^2 coefficient of 0.11. The extra work is the parity — a corner is the bishops' own colour, so the king has to be driven to an edge square of the other one.



8 × 8 deepest, **31** plies



Kc1 Bb1 Bd1 vs ka1
the corner net



Kf1 Bb1 Bd1 vs ka2
lost in **24**

Left, the deepest win, ending 16. Bc6–b7 stalemate on a7: b7 is a bishop the king defends, a6 and a8 are the other two white squares and the bishops hold both, b6 and b8 are dark and the king holds them. **Centre**, the same net in the corner and the reason one is possible at all — a bishop *occupies* b1 rather than attacking it. **Right**, why this table converts: Black is lost in 24, and his ...ka2 × b1 reaches K + B vs K, which §4.3.4 shows White wins in 17. Score that capture a draw, as the ordinary invariant does, and Black holds it.

4.3.4 The lone bishop

Delete one bishop and the endgame survives, which under ordinary rules it does not: K + B vs K is a dead draw with nothing to tabulate. Under the capture rules a lone bishop and king can take a bare king's last square away. Depths below are capture plies.

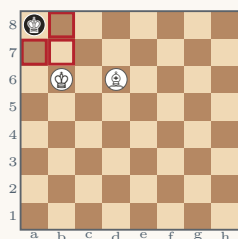
n	4	5	6	7	8	10	12	14	16
K + B wins	1 660	8 424	29 280	8 636	7 360	11 168	15 744	21 088	27 200
share	98.6 %	100 %	100 %	10.6 %	3.81 %	1.40 %	0.63 %	0.32 %	0.18 %
deepest (plies)	17	29	41	33	19	19	19	19	19
K + N wins, for comparison	1 764	8 792	30 824	3 180	2 844	2 900	2 900	2 900	2 900

One bishop beats one knight, and range is the reason

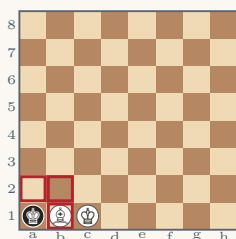
The two pieces break at the same place, between 6×6 and 7×7 , and after that behave nothing alike. **Before the break the bishop is perfect and the knight is not:** on 5×5 and 6×6 K + B wins from every legal placement, and the knight never reaches 100% on any board. **After it the bishop's win set keeps growing and the knight's does not:** from 8×8 to 16×16 the count is exactly $96n^2 + 176n - 192$ for even n and $96n^2 + 176n - 220$ for odd, against the knight's flat 2 900, so the share falls as n^{-4} rather than n^{-6} .

The reason is where the losing king may stand. For the knight, 24 squares, a staircase in each corner; for the bishop the whole rim, every border square plus b2, g2, b7, g7, which is $4n$ and grows with the board. A bishop can cover an escape square from the far end of a diagonal, so each spot on the edge has order n bishop squares that do the job where a knight must stand next door. That $4n \times O(n)$ is the quadratic. What the bishop pays is depth: frozen at 19 plies from 8×8 up, against the knight's 29.

The finishes are counted exactly, and the colour rule is absolute. There are $20n - 24$ stalemate positions on every board from 5×5 to 16×16 against the knight's flat 40, and in every one the black king stands on the border ring and **the bishop is on the opposite colour to the square the king is stalemated on** — 136 of 136 at 8×8 , 216 of 216 at 12×12 . With two bishops the constraint was that they complement each other; with one, the corner.



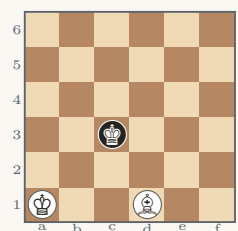
Kb6 Bd6 vs ka8: a7, b7 to the king, b8 to the bishop, a8 attacked by nothing
lost in 2



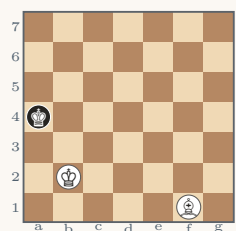
Kc1 Bb1 vs ka1: the bishop is itself one of the three squares, and its own king defends it
lost in 2

Both finishes are the same shape: the king takes two of the corner's three neighbours and the bishop the third. The two the king cannot cover together are of the colour opposite the corner, and a bishop can only take a square of its own colour, so it is forced, the same way on every board.

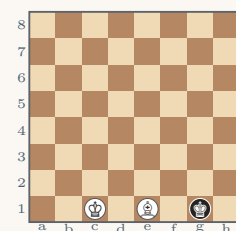
The longest win on each board, either side of the break. The deepest position gets *shorter* as the board grows through it and much harder to find: before the break White is spoilt for choice, after it he generally has one move.



6 × 6: **41 plies**, 104 placements share it, not one forced



7 × 7: **33 plies**, 24 placements share it, all 24 forced



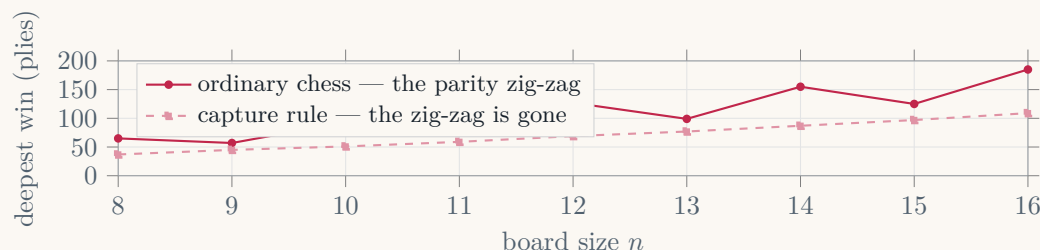
8 × 8: **19 plies**, 288 placements share it, 224 forced

Each line stops at the stalemate; the two plies that take the quoted depth past it are Black's forced step onto a covered square and the capture. After 20. Be4 at 6 × 6 all five of f4's squares are gone: e3 and e5 to the king, f3 and f5 to the bishop, e4 to the bishop with its own king behind it. Moves marked ! have no alternative. K + B vs K is not one of the eight endgames the program builds, so these numbers come from the rules-first solver of §6.3 with its generator changed, and from a second solver written win-and-draw only; the two agree placement for placement from 5 × 5 to 10 × 10.

5 King, bishop and knight

The hardest of the bare-king endgames: the only one whose depth is quadratic rather than linear, and the only one whose limit is the one-byte entry rather than memory.

5.1 Growth



The parity zig-zag is a mating artefact, and the capture rule deletes it

Ordinary KBNK alternates violently with the parity of the board, every even board far deeper than the odd one above it, because the mate needs a corner the bishop can reach and on an odd board all four corners are one colour (§5.2.3). Under the capture rule **the alternation is simply gone**. A quadratic fitted to the chess figures has residuals of 17.65 plies, no smooth curve being able to follow a zig-zag; fitted to the capture figures, **0.56**, with an n^2 coefficient of 0.31. The same eighteen numbers go from unfittable to one of the cleanest fits here.

The natural reading is that the zig-zag was never about the chase and always about the finish: if a

stalemate ends the game in White's favour, White need not deliver his king to a corner of one particular colour. That sentence is an interpretation of measured figures, and §5.2.3 is its evidence.

5.2 Normal chess

5.2.1 Deepest mate

KBNK is the only one that bends

Measure each endgame's slope over the first and last third of its own computed range. A linear law keeps the same slope.

	range	early slope	late slope
KQK	8–60	1.45	1.48
KRK	8–50	2.36	2.36
KBBK	8–22	2.60	2.80
KBNK (odd)	9–21	5.25	7.00

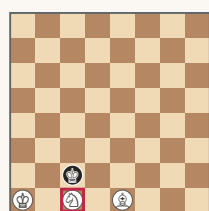
And there is a reason. A queen, rook or bishop repositions anywhere in one move, so a repositioning costs $O(1)$ and the depth stays linear. A *knight cannot*: crossing an n -wide board takes $\Theta(n)$ knight moves, and the mating procedure needs $\Theta(n)$ repositionings. So KBNK should grow like n^2 , and it does — over the odd boards $0.1411n^2 + 2.095n - 0.82$ holds to a maximum residual of **0.64 moves** at all ten points, and out of sample: fitting only $n \leq 19$ predicts 106.0 at $n = 21$ against a measured **105**.

5.2.2 Examples

A single fixed position, unchanged for all n

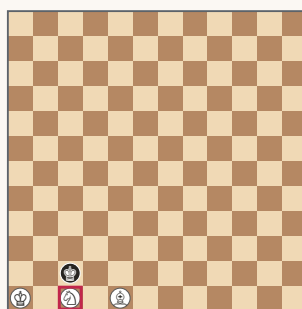
wK a1, N c1, B e1, bK c2, White to move

is a deepest KBNK win on **every board from $n = 8$ to $n = 19$** — twelve consecutive sizes, tables from 1.9 million to 2.1 billion entries per side, depths from 33 to 112 moves. It does not scale with n at all.



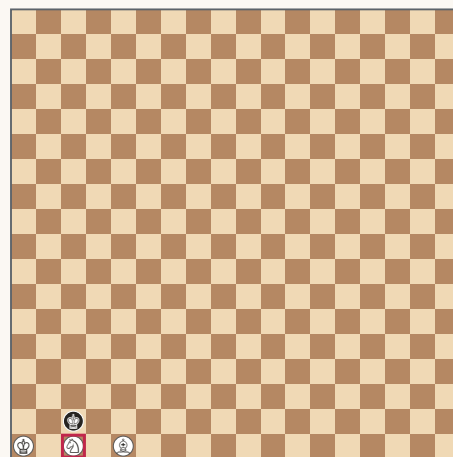
○ N

$n = 8$ — mate in 33
138 classes



○ N

$n = 12$ — mate in 64
1 class (unique)



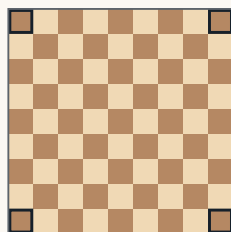
○ N

$n = 18$ — mate in 112
9 classes

Drawn at a common *square* size and aligned on a1: the board grows, the position does not move. The ringed knight is the hanging one.

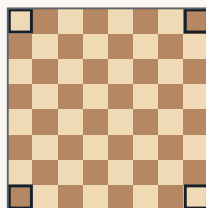
Why this is the worst case. The bishop on e1 matches a1, so a1 is a mating corner and the black king already stands beside it — but the knight on c1 hangs, the white king two files away and a bishop unable to defend along a rank. White spends a tempo saving it, the black king escapes, and he must be driven the length of the board and back, the line mating **on a1 itself** with the knight returned to c1. Everything a larger board adds is escape-and-redrive distance, which is why the depth grows quadratically while the position stays fixed.

5.2.3 Where square colour decides the game



[N]

$n = 9$ — all four corners *dark*



[N]

$n = 8$ — two dark, two light

On an odd board, with the bishop **off** the corner colour, White wins nothing that was not already mate in two — checked for $n = 3, 5, 7, 9, 11, 13$, where at $n = 13$ that is 328 839 036 of 328 844 528 legal white-to-move placements drawn. The question only makes sense for odd n : square colour is a D_4 invariant exactly then, mirroring the files on an even board exchanging the colours.

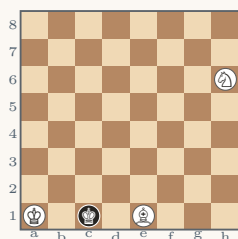
That is what makes the depth curve zig-zag. On an even board two corners are the wrong colour and the expensive part is marching the king out of one and along the edge to a right one; on an odd board, when the bishop matches at all, every corner is a mating corner. Hence 93 at $n = 16$ against 63 and 76 either side.

5.3 Capture the king

5.3.1 Deepest win

37 plies on 8×8 against 65 for chess, and the whole difference is in where the knight starts.

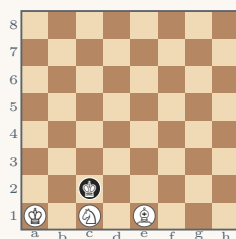
5.3.2 Examples



○

[N]

chess: $Ka1\ Be1\ Nh6\ vs\ kc1$
65 plies, the hardest mate in this document



○

[C]

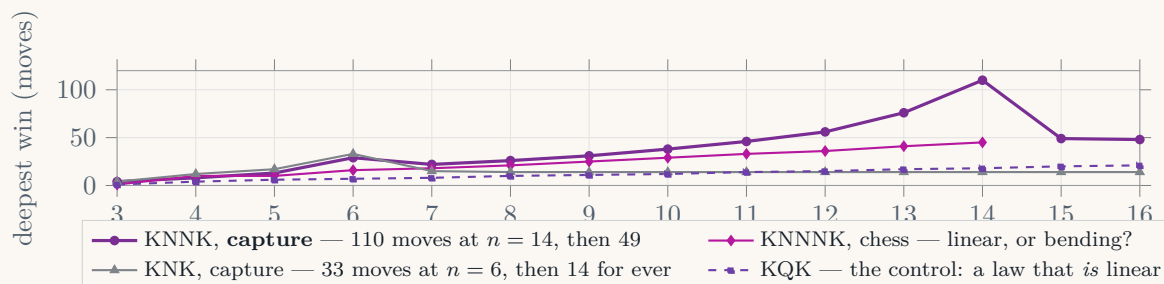
capture: $Ka1\ Be1\ Nc1\ vs\ kc2$
37 plies

The knight has come home, starting beside the bishop instead of across the board: the long journey in the chess line is the knight fetching the black king out of the wrong corner, and under this rule there is no wrong corner.

6 Knights

Every table so far rests on one sentence: whatever Black captures cannot mate, so the capture ends the game in a draw and lies outside the table. §7 breaks that sentence by arming Black; this section breaks it from the other side, Black staying bare to the last move while it is *White's* remaining material that stops being harmless.

6.1 Growth



Two of the four end in a cliff, and the shape before it is the same both times: a depth climbing faster and faster right up to the board where the win disappears — KNK at $n = 6$, KNNK under the capture rule at $n = 14$. Against those KQK is boringly straight, and KNNNK cannot yet be told apart from it.

6.2 Normal chess

6.2.1 Deepest mate

n	4	5	6	7	8	9	10	11	12	13	14
K+N+N+N vs K, deepest (plies)	19	19	31	35	41	49	57	65	71	81	89
K+N+N vs K, deepest (plies)	1	1	1	1	1	1	1	1	1	1	1
K+N vs K	no mate exists on any board										

One knight has no table. Built from 3×3 to 24×24 , K+N vs K holds zero mates every time — not no *forced* mate, but no placement in which Black is mated at all. Its only other terminal is stalemate, and there are exactly 40 of those on every board from 4×4 up, ten per corner. **Two knights have a table one ply deep.** **Three knights grow linearly**, at 8.0 plies a board over $8 \leq n \leq 14$; mate in 21 on the ordinary board, with no published value to check it against.

K + N + N vs K: mates exist; zugzwang does not

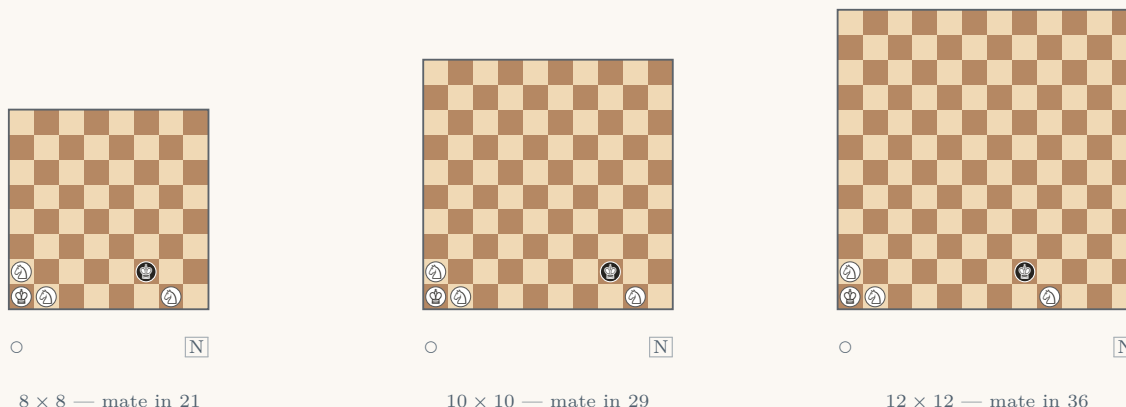
The textbook fact is that two knights cannot *force* mate. It is easy to hear that as “two knights cannot mate”, and the difference puts this endgame on opposite sides of §1’s invariant.

On 8×8 there are **120** placements in which a lone black king is mated by king and two knights, exactly $16n - 8$ on every board from 4×4 up. And yet the deepest white win is **one ply** from 3×3 to 16×16 : **White wins only where he mates on the move**, and such placements number $104n - 216$ for $n \geq 5$, one in a million and a half at 16×16 . Mates exist; zugzwang does not, so **Black is never obliged to walk into one**, and black-to-move losses and mates are the same number on every board.

With one exception, the smallest board. On 3×3 there are 36 black-to-move losses against 32 mates: one symmetry class in which Black has moves and all of them step into mate, there being nowhere else to stand.

6.2.2 Examples

The deepest three-knight win has the same shape on every board from 8×8 up, and it is not the black king in a corner. It is the *white* king in one, two of his knights on the squares he would use to get out, the third far off on the first rank with the black king beside it.

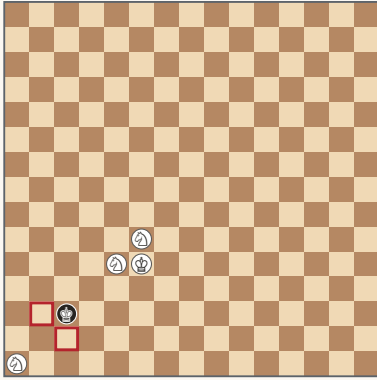


The worst case is not a resource of Black's: it is White untangling his own pieces before the hunt can start, and then walking the whole board. On 13×13 and 14×14 it is not merely the same shape but the same five squares, wK a1, wN b1, wN a2, wN l1, bK k2, at 81 plies and 89.

And the win is not universal, which no other bare-king endgame here can say: one placement in eighty is drawn at 8×8 . The obvious hypothesis is that the draws are where a knight hangs, and unlike §7's it survives measurement.

n	drawn			won		
	loose	guarded	quiet	loose	guarded	quiet
8	94.9 %	0.0 %	5.1 %	23.6 %	5.8 %	70.6 %
10	90.6 %	0.0 %	9.4 %	17.4 %	2.7 %	80.0 %
12	88.5 %	0.0 %	11.5 %	13.1 %	1.4 %	85.6 %

Loose means a knight on a square the black king attacks with nothing defending it; *guarded*, attacked but defended; *quiet*, the black king attacks nothing. A hanging knight is present in nine drawn positions in ten and one won position in eight, and the gap widens with the board. Two more things fall out: the **guarded** column is not merely small but *constant*, exactly 128 placements from 7×7 to 12×12 , and the **quiet** draws climb the whole way, 0.1 % of the draws at $n = 5$ to 11.5 % at $n = 12$.



○ N
wK f5, wN a1, wN e5, wN f6, bK c3 on 15×15 —
drawn

One of the quiet draws, on the largest board built: nothing is attacked and every white man is within three squares of the black king.

The a1 knight is trapped. Its two moves, b3 and c2, are both ringed, attacked by the black king and defended by nothing; and leaving it does not help, the black king being **two** king-moves from a1 and the white king **five**. So a knight goes, and what is left is K + N + N vs K, won only where White mates on the move. That is why the quiet draws are a *share* that grows while the draws get rarer: a trapped corner knight is a local accident, so these positions grow far more slowly than positions do, but among the draws they come to dominate the hanging-knight cases, which need Black next to a knight already.

6.2.3 What that costs the table above it

The chain is three deep and only the top link costs anything. . . K×N leaves bare kings out of KNK and KNK out of KNNK, both dead draws; out of KNNK it leaves KNNK, which is **sometimes won**, so KNNK must look the answer up — the second endgame here that converts, after KQKR, and the first whose conversion runs one way only. The cost is nothing (520 MB for KNNK at 16×16 , against 41 GiB for KNNK). The index gains one thing: three like knights are an *unordered triple*, one of $\binom{n^2}{3}$ rather than n^6 .

6.3 Capture the king

6.3.1 Deepest win

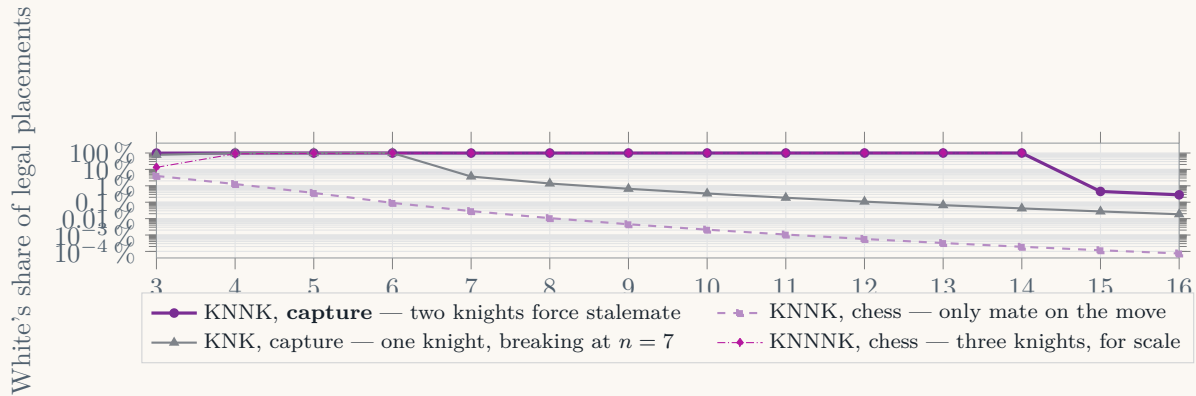
§10.1's theorem makes the capture rules chess with two clauses replaced, and the second cannot fire here: a cornered king has three on-board neighbours and two knights can block at most two. So **the capture rules are exactly chess with stalemate scored as a loss** — one clause, worth the whole endgame, because two knights cannot force mate but they *can* force stalemate.

One clause, and a drawn endgame becomes a won one

n	4	6	8	12	14	15	16
chess: White wins	120	408	616	1 032	1 240	1 344	1 448
share	1.28 %	0.09 %	0.011 %	0.0006 %	0.0002 %	0.0001 %	0.00008 %
capture: White wins	9 348	447 832	5 713 976	179 381 448	644 455 884	5 162 568	5 461 920
share	100 %	100 %	99.38 %	99.85 %	99.92 %	0.45 %	0.28 %
deepest (plies)	15	57	51	111	219	97	95

On 8×8 the same men go from **616** won placements to **5 713 976**, a factor of 9 276, and up to 6×6 White wins *every* legal position. Two figures say the rule does what it claims and no more: the **mate count is identical under both rules**, $16n - 8$ either way, and the **chess stalemate count equals the capture-rule loss count exactly**, 3 864 at 8×8 . The clause relabels precisely that set — and then it goes, the deepest win running 51, 61, 75, 91, 111, 151, **219** plies for $n = 8 \dots 14$ before the win collapses to 0.45 % at 15×15 .

Both transitions on one axis, the ordinary rule for comparison. The scale is logarithmic, the two rules being six orders of magnitude apart; colour is the material and line style the rule.



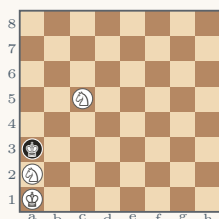
Three knights under ordinary rules and two under the capture rules sit almost on top of each other up to 14×14 — an extra knight and the changed clause are worth about the same — and then the capture line falls off a cliff the KNNK line never reaches. What is left of the lone knight is not a share but a fixed count, 2 900 wins against n^6 placements, which is why its line decays as n^{-6} .

K + N vs K under capture rules

n	4	5	6	7	8	10	12	14	16
White wins	1 764	8 792	30 824	3 180	2 844	2 900	2 900	2 900	2 900
share	98.9 %	98.9 %	99.5 %	3.7 %	1.4 %	0.34 %	0.11 %	0.04 %	0.02 %
deepest (plies)	23	33	65	29	27	27	27	27	27

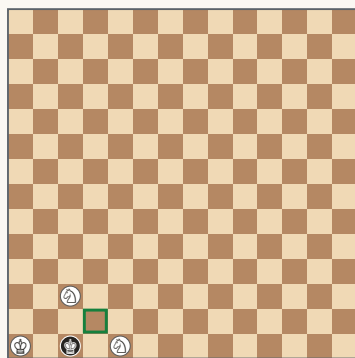
The knight below does the same thing, sooner, then collapses at $n = 7$ to a fixed handful: **2 900 placements on every board from 10×10 up**, never deeper than 27 plies, with exactly 40 stalemates whatever the size. That is why the KNNK capture table converts into a KNK capture table rather than scoring $\dots K \times N$ as a draw: it is not one.

6.3.2 Examples



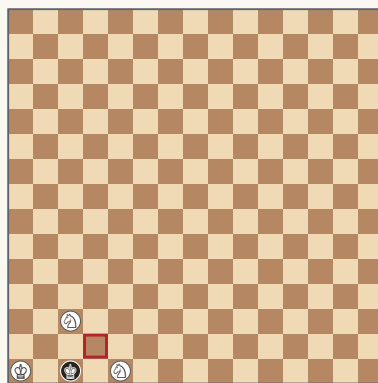
wK a1, wN a2, wN c5, bK a3 — a draw in chess, a loss under capture rules

The relabelled position: Black is not in check, has no move, and every square he would like is covered, a2 by the king. In chess that is stalemate and a draw; here it is a loss.



○ [C] ○

14 × 14 — White wins, 104 moves



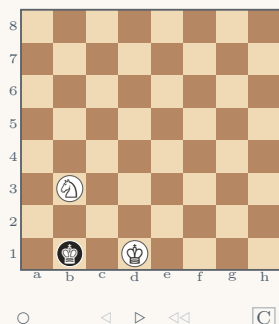
○ [C] ○

15 × 15 — drawn

The same four men on the same four squares, the only difference one extra rank and file, and nothing attacked anywhere. The black king has five squares and **four are covered** — b1 and d1 by the knight on c3, b2 by the white king, c2 by the knight on e1 — and under these rules stepping onto a covered square is legal and loses at once. Only the ringed d2 is free.

On 14 × 14 that is enough: Black runs out through d2 and White spends 104 moves putting him back. On 15 × 15 it is not. This one placement carries the whole runaway — **52 moves at 12 × 12, 71 at 13 × 13, 104 at 14 × 14, then drawn at 15 and 16** — and the draw is bought not by a threat or a tempo but by the extra rank.

The longest lone-knight win, and its only move. The 2900 survivors are not scattered: from 8 × 8 up the losing black king stands on exactly **24 squares**, a six-square staircase in each corner (a1, b1, c1, a2, b2, a3 and its images), the same set on 8 × 8 as on 12 × 12. That is why the count freezes rather than growing.



○ < > << [C]

Kd1 Nb3 vs kb1 — mate in 14, 27 plies

The deepest survivor, stepped to the capture. Moves marked ! have no alternative: every other legal move draws. Black is not running — he is already caught, and shuffles a1–a2–a3–b1 throughout. Almost the whole length of the win is the knight going round twice, c5–d3–b2–c4 and then c4–d6–b5–c3.

The first move is unique, and it looks wrong. Eight of White's nine legal moves draw. Nc5 takes the knight *away*: on b3 it already eyes a1 and c1, and c5 attacks nothing within two squares of the black king. Ten of eleven draw at move 11 likewise. **And c5 is the knight's one square wherever it starts:** all 72 placements sharing the maximum depth on 8 × 8 have a unique winning move, and the seven that are this same Kd1 against kb1 — knight on b3, a4, e4, a6, e6, b7 or d7, precisely a knight's move from c5 — all play Nc5. What ends it is not a mate, K + N having none: after 14.Nc3 the king on a1 has three squares, all covered.

How this one was checked, and what it caught

The unreduced solver disagreed with the table, and it was right. Phase B's two-phase alternation assumes a black-to-move loss always has value $d + 2$; a *conversion* can name any depth, and KNNK under capture rules reaches 65 plies, so values were written that the alternation never came back to retract. KNNK escaped this only because the KNNK table beneath it is one ply deep. The fix is the bucketed induction §10 needed for the same reason.

Behind the numbers: Bellman re-derivation for $10 \leq n \leq 16$, both sides of the transition included; the unreduced solver over every ordered placement for $n \leq 10$; and a third solver written from the rules of §10 rather than from the theorem, reproducing the table at $n = 4$ and $n = 5$ with every depth larger by **exactly two plies**. The lone knight has been through that third solver on every board to 14×14 , and the line above checked position by position, all 28 of them, each a ply shorter than the last.

7 Queen against king and rook

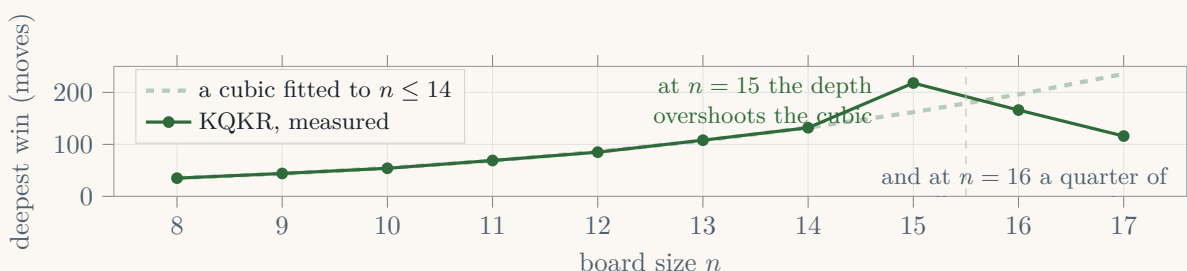
Everything up to here rests on Black having a lone king, which is what makes the tables cheap. This section crosses that line and gives Black a rook.

Three things KQKR needs that none of the others do

Signed entries, $\dots R \times Q$ leaving a bare white king against king and rook, which White *loses*, so a white-to-move entry can now be a loss. **Conversion**: $Q \times R$ leaves KQK and $\dots R \times Q$ leaves KRK with the colours swapped, both read forwards out of tables this program already builds — and still no un-capture generator, a predecessor under a capture having *five* men. **A four-way induction**, either side being able to win, so every ply runs a retraction and a forward re-test for each.

One fact keeps the bookkeeping small: **no capture is ever bad for the side making it**, taking the rook leaving a white win or draw and taking the queen a black win or draw, never a loss. So a converting move can seed a win but never has to be scheduled as a loss at a particular depth.

7.1 Growth



There is no capture-rule line on this chart, and §7.3 says why.

7.2 Normal chess

7.2.1 Deepest mate

Mate in 35 on the ordinary board is the published figure, not fitted to. What matters is the rate: between boards two apart the local exponent α for which $d \propto n^\alpha$ runs 1.46, 1.94, 2.49, 2.86, 4.91 from $6 \rightarrow 8$ to $13 \rightarrow 15$, climbing straight through 2 and onward. A cubic fitted to $n \leq 14$ has residuals under a move and predicts 162 at $n = 15$,

$$d(n) \approx \frac{1}{18}n^3 - \frac{1}{7}n^2 - \frac{169}{126}n + \frac{559}{21},$$

but it is an interpolant: over so short a range the n^3 coefficient is not identifiable, sliding from +0.08 to -0.02 as the window moves by one board with no loss of fit. **The exponent does not converge to 3. It passes through it.**

And then the win disappears

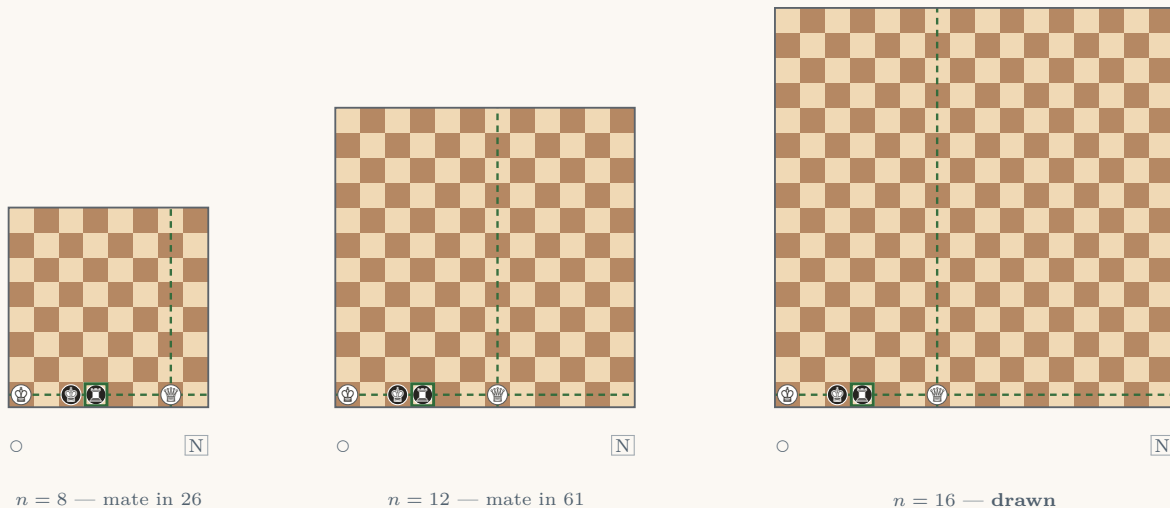
At $n = 15$ the depth overshoots the cubic by more than fifty moves. At $n = 16$ the drawn fraction, falling steadily for seven board sizes, jumps from one position in eight hundred to more than one in four, and the deepest win *falls*. A whole class has stopped being winnable, and the class is visible in the depth histogram, which is bimodal at every board size:

n	10	11	12	13	14	15	16
shallow mode (plies)	21	23	25	27	31	33	35
deep mode (plies)	85	111	141	183	227	395	—

The shallow mode is untroubled all the way across; the deep one accelerates and then, at $n = 16$, is simply absent. 65 million of the 241 million wins at $n = 15$ run deeper than 150 plies, against 1.7 million at $n = 16$, and the 110 million drawn positions are what that mode used to weigh.

7.2.2 Examples

The cleanest evidence is one placement, taken from the drawn list at $n = 16$ and probed on every board that holds it: all four men in the corner, the rook on the queen's rank between her and the black king.



Dashed, the queen's rank and file; ringed, the interposing rook. The position never changes, only the empty space behind it — and the extra room is room to keep interposing.

n	8	9	10	11	12	13	14	15	16
wK a1, wQ g1, bK c1, bR d1	26	30	37	51	61	77	95	176	draw

This is the shape of a phase transition: a quantity diverging as a parameter approaches a critical value, and past it the thing it measured is gone.

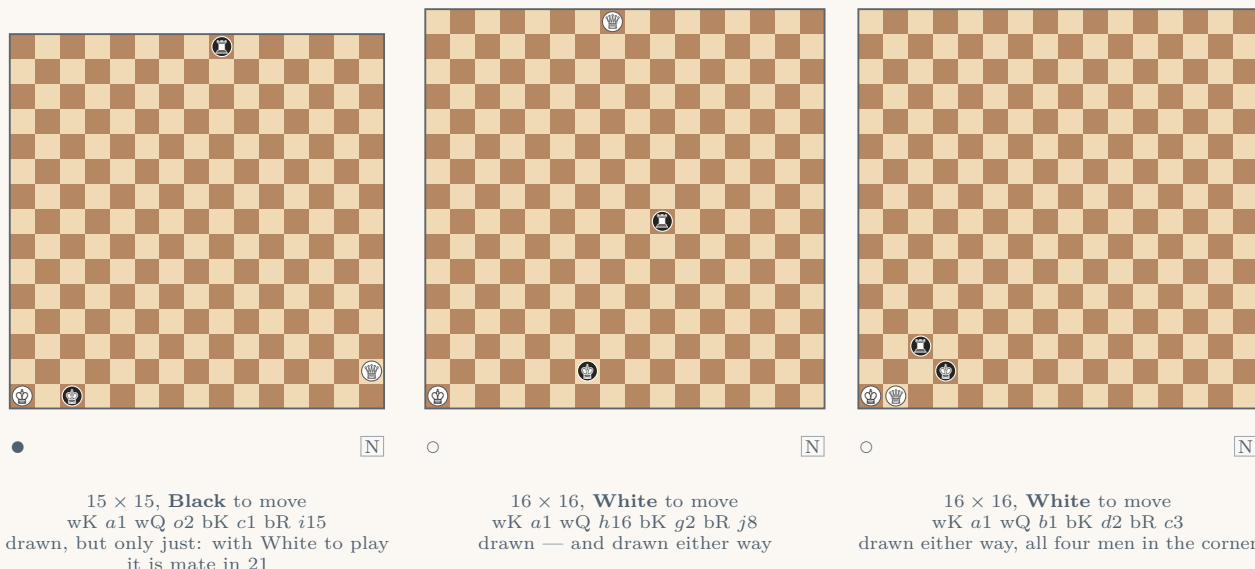
7.2.3 What the new draws look like: nothing in the air

A position can be drawn *tactically* — Black checking, hitting the queen, a king in contact — or *structurally*, nothing attacked and White with no way in. The two are told apart mechanically: call a placement **quiet** when nothing whatever is attacked.

Below the transition every draw White faces is tactical; above it most are not

drawn symmetry classes	8×8	15×15	16×16
White to move	8 963	304 483	110 091 285
of them quiet	0	0	76 718 643 (69.7%)
Black to move	78 518	4 573 792	338 880 229
of them quiet	35 284 (44.9%)	3 145 919 (68.8%)	276 036 657 (81.5%)

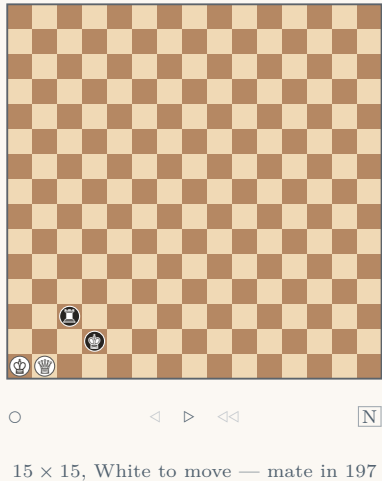
Quiet draws with *Black* to move exist at every board size, so the test is not vacuous. But with **White** to move, up to and including 15×15 , there is **not one**: 287 230 of the 304 483, 94 %, have the rook *checking*, and the rest have it hitting the queen, the queen hitting it, or a king in contact. At 16×16 seven of every ten do not.



None of the three has anything attacked. The 15 × 15 position is drawn *only* because it is Black’s move; the two at 16 × 16 are drawn whichever side is to move, which cannot happen on any smaller board. At 15 × 15 Black holds 3.1 million quiet positions, but only *with the move*.

7.2.4 What the win is, and what one more file does to it

On 8 × 8 the KQKR win is *local*: the white king walks to the black one and zugzwang wins the rook. On 15 × 15 it is not local at all.



Black’s first move is Rc3–c15: the rook bolts twelve squares up the file, as far from everything as the board allows, and that is the whole defence. Over the 383 plies before it falls Black makes **81 rook moves averaging 5.7 squares**, 28 of them eight or longer. For **68 %** of the line the rook stands six or more squares from its own king, that distance *growing* as the game goes on (6.3, 8.1, 10.3, 10.4 by quarter), and for the same 68 % the black king is within one square of an edge.

So the rook is not sheltering beside its king, where a fork would win it: it is parked far off on an open line, too distant to be forked with the king and still able to interpose or check. White’s king must cross the whole board, and his only resource is to take safe squares away from the rook until it runs out. **The number of safe squares is what the board size controls.**

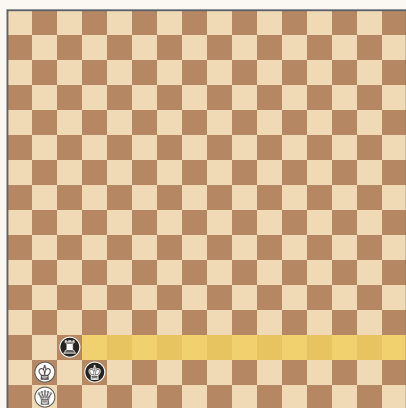
The same rook moves: every one loses at 15, every one draws at 16

After 1. Kb2, Black shuffles the rook along the third rank. Each entry is the value of the position after that rook move.

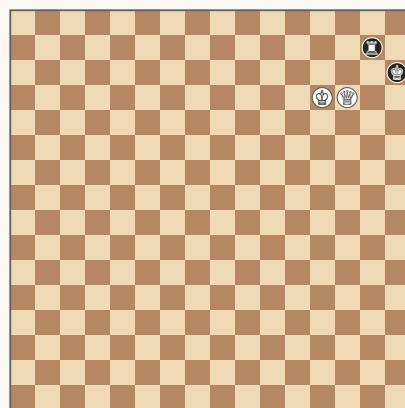
rook to	d3	e3	f3	g3	h3	j3	l3	o3	p3
15 × 15	W 194	W 189	W 195	W 193	W 188	W 188	W 188	W 188	—
16 × 16	draw	draw	draw	draw	draw	draw	draw	draw	draw

Note the saturation at 15 × 15: from h3 on the value stops moving at mate in 188, so extra distance buys Black nothing more *there* and White still grinds it out. One file wider the whole family holds — not

because Black gains a square ($p3$), but because the scheme becomes tenable, 21 replies flipping from lost to drawn at once.



● N



○ N

16×16 , after 1.Kb2: every shaded square holds the draw. On 15×15 the same rook moves all lose.

16×16 , the identical shape pushed into the far corner — **mate in 33**. Take the room away and the win returns.

The second board is the control. Slide the four men diagonally without changing their shape and the value tracks only the open board left behind Black: at 15×15 197, 198, 177, 149, 100, 29 for shifts 0, 4, 8, 9, 10, 11, and at 16×16 drawn for all of those, then **mate in 33** at shift 12. The win comes back the moment Black is jammed against $p16$. The operative variable is the room.

The class that flips is one class, and it can be recognised

The set *won but deeper than 150 ply* at 15×15 and the set *drawn* at 16×16 agree on every measure taken: 65 203 653 against 110 091 285 positions, 27.0 % of the legal ones against 26.6 %, 69.3 % quiet against 69.7 %, rook within three of its own king in 39.5 % against 36.6 %. Same signature: a compact formation, rook near its king, board to spare. The deep wins that *survive* at 16×16 , only 1 689 276, are the mirror image, Black's king already *on* an edge in 50.7 % of them. Hence the drawn fraction barely moving to 17×17 : the criterion is local, so once there is board enough the share stops growing.

The corner placement's depth on every board that holds it — 30, 36, 43, 56, 72, 93, 114, **197**, then **drawn** for $n = 8 \dots 16$ — fits $d(n) \approx C/(n_c - n)$ with $n_c \approx 16.4$ from the closest pair (14, 15) and 16.7–16.8 from (12, 15) and (13, 15).

What this does and does not settle

Does. Growth far steeper than quadratic is real, and giving the defender a piece is what produces it. With a bare black king the confinement region is described by a handful of numbers of size $O(n)$ and re-forming it costs $O(n)$, capping the product at n^2 ; a rook that can interpose on the queen's line breaks that ceiling.

Does not. There is no growth *law* here, cubic or otherwise: the local exponent rises across the whole measured range instead of settling, so “cubic” names the board size at which it passes 3 and nothing more. Whether any fixed material is cubic *asymptotically* is open, and this suggests why it may be hard: the defensive freedom that lengthens the win eventually saves it outright. The transition is bracketed between $n = 15$ and $n = 16$ and no more — not shown to be sharp, nothing beyond $n = 17$ computed.

7.3 Capture the king

7.3.1 Deepest win

The one gap in this document

There is no capture-rule table for K+Q vs K+R here. Every other material has one; the reason is mechanical. `src/kqkr.cpp` is a separate solver with signed entries and a four-way induction, and it does not carry the `stalemate-loss` flag the bare-king solver does. Adding it is a real change to delicate code, and it has not been made.

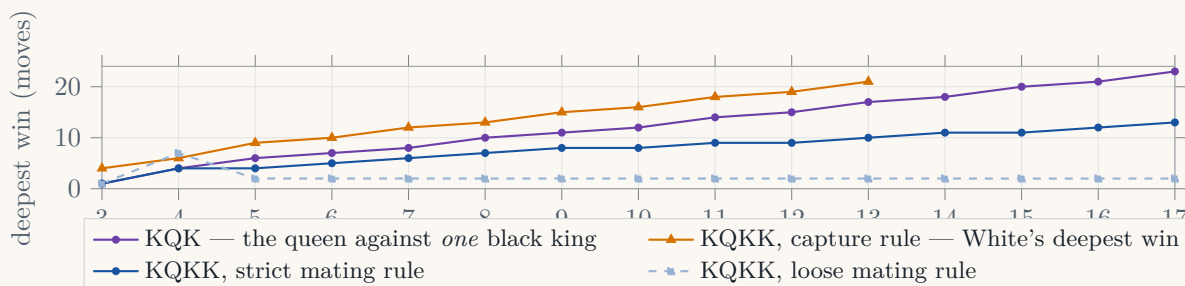
§10.1's theorem bounds how much could change without any table being built. Both sides have exactly one king, so the two games differ *only* where the side to move has no legal chess move: every other value in §7.2, the transition at 16×16 included, carries over unchanged. The missing table would answer a narrow question: how many KQKR stalemates are there, and what does refunding them buy? On the materials that *were*

computed the answer is “a little, then possibly a lot” — the queen gained one move at every size, the rook nothing below 9×9 , and two bishops eventually *lost* twelve plies to new and harder wins. KQKR is the one endgame here where Black has a piece to be stalemated *with*.

8 Queen against two kings

Black’s second man is a king, which puts this material outside ordinary chess. §10.2 defines the two *mating* rules and §10 the capture rule; this section gives the numbers for all three.

8.1 Growth



Four lines, four different games. The **strict** rule is the only one easier than ordinary KQK, at every size from 5 up: two kings that must both be kept out of check are a liability. The **loose** rule flattens to mate in 2 from $n = 5$ on, its wins a fixed handful of corner cages. The **capture** rule is the only one running *above* KQK, White having to win a king and then mate with what is left — and it understates that game, being White’s half: Black’s deepest win grows from 18 moves at $n = 8$ to **51** at $n = 13$.

8.2 The mating rules

8.2.1 Deepest mate

Strict: everything is won, and sooner than against one king

Every legal white-to-move position is a win, at every board size from 3 to 17: the white-to-move draw count is exactly zero at each. The 8×8 table holds 2 534 392 legal white-to-move placements, wins all of them, and the deepest takes **7** moves against **10** for the same queen against a single king.

The second king is a liability because Black must keep *both* out of check with *one* move, and a queen attacking both at once cannot be answered. The method is therefore a **fork** rather than a corner, and a fork has no use for the edge — which is the whole of the gap in the chart, driving a king to the rim costing a distance that grows with n while arranging a fork does not. Of the 307 604 mating placements on 8×8 , only 20.3 % have both kings on the rim, and in **56.8 %** the white king does nothing at all; on 12×12 , 78.8 %.

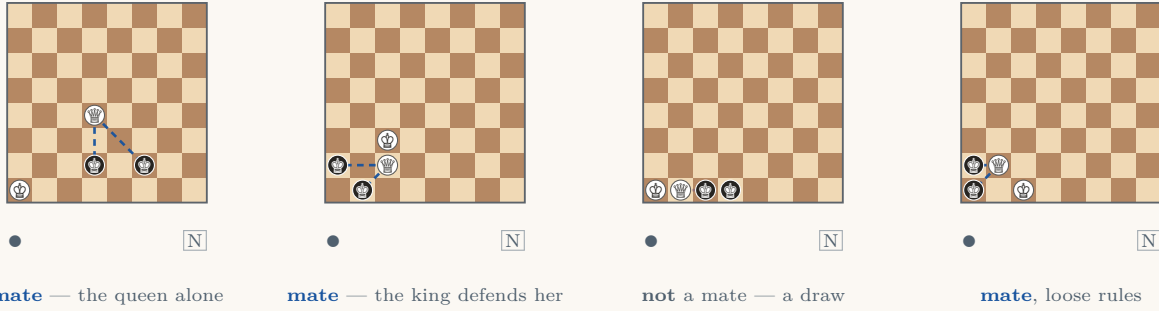
Loose: the same men, and the endgame vanishes. Let a king stand in check, forbidding only both at once, and almost nothing is left. On 8×8 **544** of 5 481 620 legal white-to-move placements are won, none deeper than mate in 2 — White never creates a win, only finishes one the placement already contains. The board holds **32** mating placements against 307 604 under the strict rule; all 32 have both kings on the rim, 87.5 % side by side, and in every one the white king stands beside a square a black king would otherwise use. The count is $80n - 96$ for $5 \leq n \leq 14$ while legal placements grow like n^8 .

What the rule costs, and there is no third kind of draw

93 508 of the 8×8 black-to-move stalemates are no move with exactly one king in check — what the ordinary rule would call mate — out of 94 624 in all. Only 1 116 are stalemates in the ordinary sense with neither king in check. At 16×16 the proportion is 99.96 %.

It costs White nothing all the same, never being obliged to walk into one, and the census says so exactly: draws = stalemates + queen *en prise* at every board size computed — on 8×8 , $94\,624 + 1\,146\,804 = 1\,241\,428$, the entire black-to-move draw count. No fortress, no positional draw, nothing Black can hold.

8.2.2 Examples



First. wQ d4 attacks d2 down the file and f2 down the diagonal; stepping either king off its ray leaves the other attacked, and the blocking squares d3 and e3 are the queen's too. The white king is in the far corner and contributes nothing. **Second.** wQ c2 forks a2 and b1; beside a black king she must be defended, and wK c3 does that while covering b2. **Third.** No legal move, c1 in check and d1 not — one king mated and not the other, which the ordinary rule calls mate and the strict rule stalemate. **Fourth.** The loose rule's mate: both kings on the rim, side by side, the white king covering the square they would use.

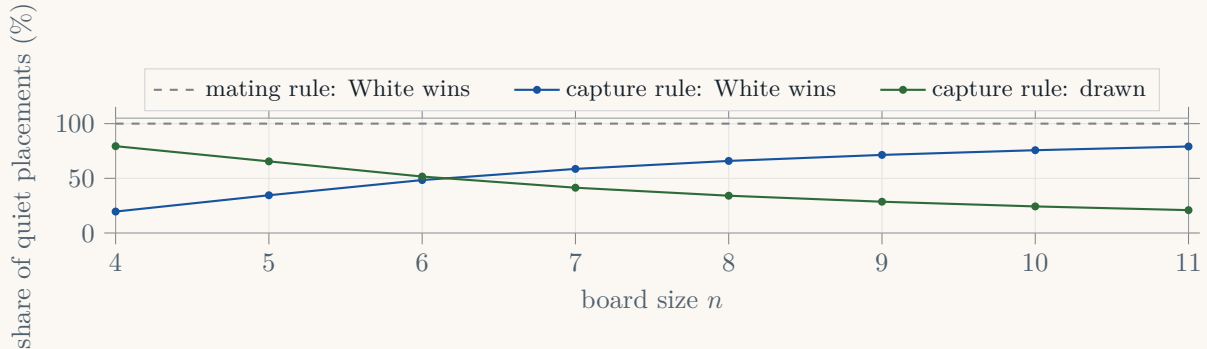
8.3 Capture the king

8.3.1 Deepest win

The same four men, read three ways. Each rule calls a different set of placements legal, so the row comparing like with like is the third, which restricts the capture table to what the strict rule admits.

8 × 8, white to move	placements	White wins	draw	Black wins	deepest
mating rule, strict	2 534 392	2 534 392 (100 %)	0	—	7 moves
mating rule, loose	5 481 620	544 (0.01 %)	5 481 076	—	2 moves
capture rule, quiet only	2 534 392	1 670 492 (65.9 %)	863 852	48	13 moves
capture rule, all placements	7 624 512	6 185 496 (81.1 %)	1 419 192	19 824	13 moves

Under the mating rules Black is lost everywhere or safe almost everywhere, depending on the reading of “both at once”. Under the capture rule he is neither: he holds a third of the board and he can win.



White's share climbs steadily with the board, the opposite of what room usually does for a defender: the more of it there is, the longer the black kings take to reach anything and the more the queen has to separate them with. Black's outright *wins* do not follow — a fixed handful of corner zugzwangs, 12 at $n = 4$ and 64 at $n = 11$, against 48 million placements.

n	3	4	5	6	7	8	9	10	11
deepest white win (moves)	4	6	9	10	12	13	15	16	18
deepest black win (moves)	3	6	10	12	15	18	24	29	36
K vs K + K, deepest (plies)	—	9	17	21	27	33	45	55	69
K + Q vs K, deepest (plies)	—	7	9	11	15	17	21	23	27

The last two rows are the two halves of every win here: White's depth is the smaller plus the manoeuvring to force a capture, Black's is dominated by the other, his win being ...K×Q and then a hunt.

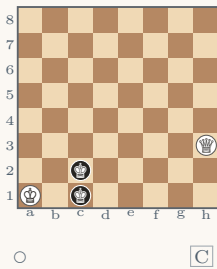
What the rules change about this endgame

Four things follow from §10's five rules, and not one of them is an extra clause. A fork wins a king rather than the game. The two black kings defend each other, so $Q \times K$ is answered by $K \times Q$ and a capture is only safe where the white king covers the square. **White can lose:** a black king beside his takes it, whatever else stands on the board. And **the queen is not expendable** — $\dots K \times Q$ leaves a white king against two black kings, which on every board a four-man table is built for is a black *win* rather than the draw it is under every other rule set here. (Past 14×14 it would be a draw; §9.2.3.)

This rule set breaks all four structural facts of §1 at once, with the same four men: Black can check, White can capture, no capture is drawn, and the table converts three ways rather than none. Hence its own solver — signed entries, three sub-endgames in dependency order, and an induction on buckets keyed by depth rather than in alternation, a conversion being able to name any depth.

8.3.2 Examples

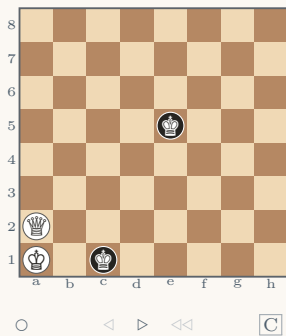
Over the same 2 534 392 quiet placements the mating rule wins 100 % with White to move; the capture rule wins 65.9 %, draws 34.1 % and loses 48, and with Black to move wins 44.9 % and draws 55.0 %. Black's third of the board is the phalanx holding: he seldom *wins* from a quiet placement, he draws by keeping his kings together and his eye on the queen. The 48 are zugzwangs, and all look like the first diagram.



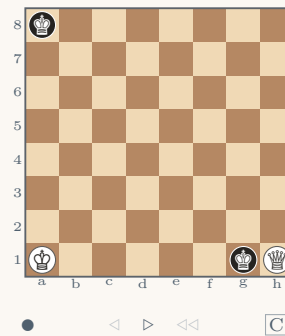
White to move, lost in 4 plies

Nothing is hanging. Kings on c1 and c2 cover a2, b1 and b2 between them and guard each other, so the white king can neither move nor take, and no queen move stops $\dots K-b2$ and $\dots K \times a1$. **White has a queen on the board and is lost because his king is sealed in the corner by two kings that defend each other** — a shape with no counterpart in ordinary chess. The tempo can decide outright: with $wK a1$, $wQ h1$ against kings g1 and a8, White to move wins in 6 and Black to move wins in 18.

Both deepest wins, played out to the capture. Neither ends in a mate: under these rules there is none.



$wK a1$, $wQ a2$ vs kings c1 and e5 — **White wins, 13 moves**



$wK a1$, $wQ h1$ vs kings g1 and a8 — **Black wins, 18 moves**

Left, 25 plies. Six moves of manoeuvring and not one check — White is not chasing either king, he is keeping them apart, and only when the d1 king is alone on its side is $Q \times d1$ safe. The second half is the $K + Q$ vs K table walking the survivor into the corner. **Right, 35 plies.** One capture and seventeen moves of pursuit: the far king walks over from a8, moves 5 to 8 forming the pair up, and from there they advance together, each covering the other. That is the whole reason the queen cannot be given away.

9 Nothing on the board but kings

Delete the queen from §10 and what is left has no pieces in it at all — **three white kings against two black ones** — the only table here in which material falls on *both* sides. Either side may take, neither capture ends anything, and (w, b) becomes $(w, b - 1)$ or $(w - 1, b)$. Six tables, built up a lattice rather than a chain:

$$(1, 1) \longrightarrow (2, 1), (1, 2) \longrightarrow (3, 1), (2, 2) \longrightarrow (3, 2).$$

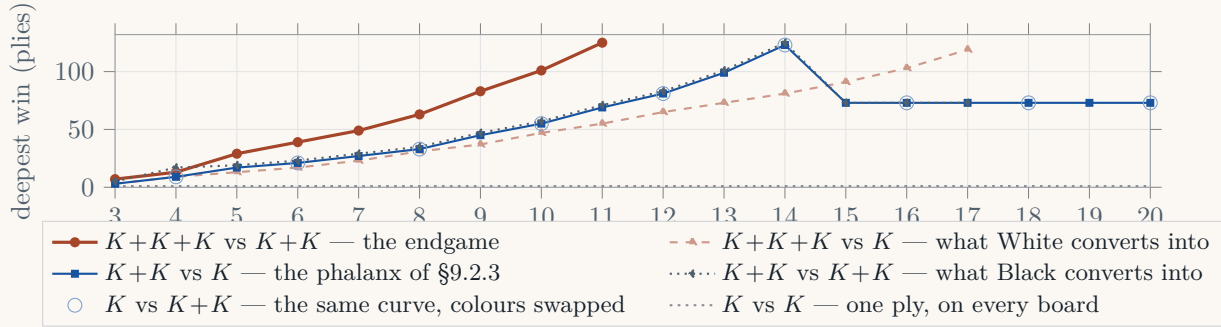
There is no other rule set here: with no piece on the board nothing gives check, so “mate” has no meaning; and rule 4 never fires, a side needing four men to be stalemated (§10.1.1) and none here having three.

Three kings beat two, and it is not close

Every white-to-move placement is a win, on every board computed, $n = 3$ to $n = 11$, up to 1 987 925 940 placements at 11×11 . Not one draw. With Black to move the endgame is a White win or a draw and nothing else from 5×5 up.

The reason is one row of the census: *up to 14×14 , three kings against one is won with **either** side to move*. So **any capture White can make wins on the spot**, the recapture not mattering, a lone king being lost against three and against two alike. No mating net, no zugzwang: White has one problem, getting a king beside a black king on his own move, and Black has one defence.

9.1 Growth



K vs K never grows at all: a bare king cannot be forced anywhere, so the position is won exactly when the two kings already touch, and then whoever is to move takes. **K vs $K+K$ is $K+K$ vs K with the colours exchanged**, and the two tables agree to the ply at all eighteen board sizes, transition included; they are built independently and from different seeds, so the open circles lying on the line are a check rather than a definition. That leaves four distinct curves.

n	8	9	10	11	12	13	14	15	16	17
$K+K$ vs K	33	45	55	69	81	99	123	73	73	73
$K+K$ vs $K+K$	35	47	57	71	83	101	125	73	73	73
$K+K+K$ vs K	31	37	47	55	65	73	81	91	103	119
$K+K+K$ vs $K+K$	63	83	101	125	—	—	—	—	—	—

9.2 Capture the king

9.2.1 Deepest win

The four grow at 14.4, 14.4, 9.5 and 20.4 plies a board, fitted from 8×8 to the last board each is won on, but none of them is linear and the slopes say little. §7.2.1's measure is the useful one: the local exponent α for which $d \propto n^\alpha$, taken between boards two apart so the odd-even wobble cancels.

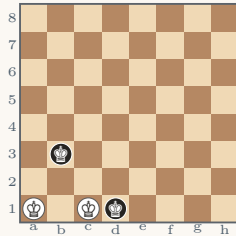
$n \rightarrow n+2$	8	9	10	11	12	13	14	15
$K+K$ vs K	2.29	2.13	2.12	2.16	2.71	—	—	—
$K+K$ vs $K+K$	2.19	2.06	2.06	2.11	2.66	—	—	—
$K+K+K$ vs K	1.86	1.98	1.78	1.69	1.43	1.54	1.80	2.14
$K+K+K$ vs $K+K$	2.12	2.04	—	—	—	—	—	—

This corrects the obvious reading of the picture. The five-man line's steps grow, +10, +10, +14, +20, +18, +24 from 5×5 to 11×11 , and a step that doubles looks like a runaway. It is not: a steady $\alpha = 2$ produces linearly growing steps by itself. What preceded the collapse of $K+K$ vs K was not its five-board plateau but the **single jump to 2.71 in the last window before it went**, and $K+K$ vs $K+K$ did the same. The five-man endgame shows no sign of it at 11×11 , and is quadratic with constant 1: d/n^2 runs 1.000, 0.984, 1.025, 1.010, 1.033 at $n = 7 \dots 11$, a cubic term comes out -0.0000 and changes sign as the window slides, and the ratio to $K+K$ vs K runs 1.909, 1.844, 1.836, 1.812 at $n = 8 \dots 11$ where a cubic over a quadratic would grow like n . It will not extrapolate: a power law over the last four boards gives 2.125, the plateau $K+K$ vs K sat on before it jumped.

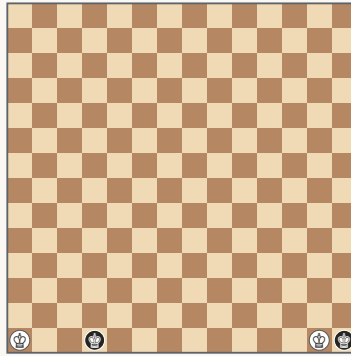
Three kings against one is the outlier, the only curve that bends twice. Its exponent *falls* from 1.98 to 1.43, then climbs back to 2.14, the turn starting at the $13 \rightarrow 15$ window — exactly where the endgame underneath it broke. Three kings do not stop beating one at 15×15 ; they begin paying for the recapture Black has just gained (§9.2.5).

9.2.2 Examples

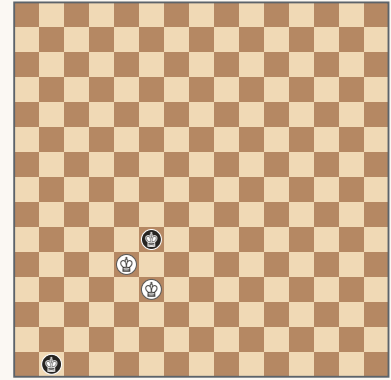
Two kings against two is exactly two plies worse than two against one, at every board from 5×5 to 14×14 : 19/17, 23/21, 29/27, 35/33, 47/45, 57/55, 71/69, 83/81, 101/99, 125/123. The first two diagrams show why — a black king left *en prise* beside a white one with its partner as far off as the board allows, so White takes for nothing, Black replies, and what remains is the deepest $K+K$ vs K hunt there is.



8×8 , **35 plies**: 1. Kc1 $\times$ d1, then 33



14×14 , **125 plies**: 1. Km1 $\times$ n1, then 123



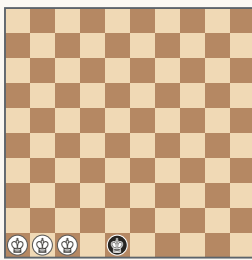
15×15 , **73 plies**: 1. Ke5 $\times$ f6, then only 72

And the identity breaks exactly where that mechanism does. The third board looks like the other two, but both endgames stand at 73 plies at 15×15 : the identity would need the hunt that follows to be worth 74 with Black to move, and it is worth **72**. The longest hunts are no longer reachable with a spare king hanging beside them.

$K+K$ vs $K+K$ dies at the same board size and far more completely — 13.3% won at 14×14 to **0.03%** at 15×15 , against 100% to 7.0% — White needing two things to go right and Black one. What survives shrinks rather than grows, 203 232, 198 748, 195 816 won placements at $n = 15, 16, 17$: corner traps with the board getting bigger around them. **Three kings catch a lone king faster than two**, from 5×5 on, and the gap widens without bound, 81 against 123 at $n = 14$. A third king is less a third pursuer than a third wall.

A coincidence, recorded rather than explained. The three-man plus four-man depth misses the five-man depth by exactly one at every board from 5×5 to 11×11 — 30/29, 38/39, 50/49, 64/63, 82/83, 102/101, **124/125** — the last a genuine prediction. It is not a law (4×4 gives 18 against 13), and the play does not support the obvious reading: were the win *force a capture, then finish*, the first capture would fall where the summands meet, and instead it comes at ply 5, 8, 17, 27, 26, 43 for $n = 3 \dots 8$ against depths 7, 13, 29, 39, 49, 63. What runs away with the board is the *herding*.

The hardest position is White's own men in his own way. As with the knights of §6.2.2: at every board from 8×8 to 17×17 the deepest $K+K+K$ vs K placement is **Ka1 Kb1 Kc1** against a black king a couple of files away.



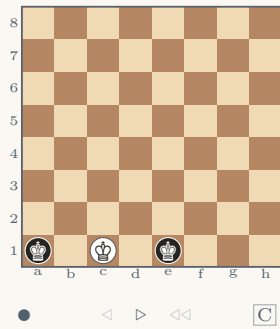
10×10 , White to move, **47 plies**

The first six moves are pure deployment: White's three kings begin jammed on three squares of one edge, each blocking the others, and his difficulty is not finding the black king. The line runs on to 24. K*i*9 $\times$ j10, the black king taken in the far corner 71 squares from where he started. That the deepest position of a completely won endgame should be one where the winning side is in its own way is a joke the tables keep making.

9.2.3 Two kings hunt down a lone king, until the board is too big

The load-bearing row of the census, computed rather than assumed. Black wins from **every one** of the 124 992 placements of K vs $K+K$ on 8×8 , at most 33 plies deep, and from every placement on every board up to 14×14 . The method is the point: the two kings stay *side by side*, so a white king taking one is taken by the other, and in that formation they walk a lone king to the edge. That is why $\dots K \times Q$ ends the game.

The other conversion runs the same way for White: **K + Q vs K** is won from every position with the move, in 17 plies against 19 for KQK.



K vs K+K, Black to move — the white king falls in 17 moves

The 8×8 win in full, 33 plies. The value falls by exactly one at every ply — +33, -32, +31, ..., +1 — the certificate that neither side has anything better.

The kings advance *abreast*, never more than a square apart, so the white king can never take one without being taken by the other: that mutual guard, not the number of men, does the work. The last two plies are the point of the whole rule set — $16 \dots Kf7-g7$ puts a black king *beside* the white one, which no mating rule here would allow, and White cannot take it because the second king covers $g7$. He has only $h8$.

The hunt has a critical board size, and it is between 14 and 15

Up to 14×14 two kings win from **every** placement with the move, and the depth climbs faster than linearly the whole way: 17, 28, 41, 50, **62** moves at $n = 8, 10, 12, 13, 14$. At 15×15 it collapses.

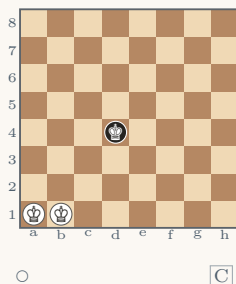
n	12	13	14	15	16	18	20	22
Black wins, black to move	100 %	100 %	100 %	7.0 %	6.0 %	4.7 %	3.8 %	3.1 %
deepest win (moves)	41	50	62	37	37	37	37	37

Seven per cent of placements are still won, and the deepest of them is 37 moves — a number that does not move again: **exactly 73 plies at every board size from 15 to 22**. Their count grows like n^4 against n^6 placements, so the share decays like n^{-2} .

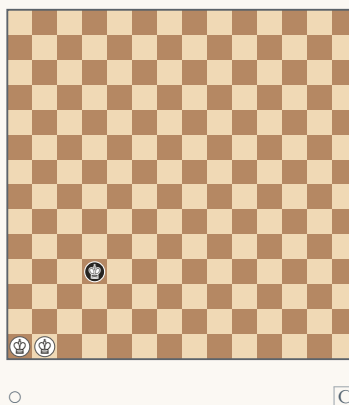
What survives is *not* a cornered white king, the first guess: at $n = 16$ his distance to the nearest edge is distributed almost identically in the two sets. Nor is it a phalanx already formed, the black kings standing adjacent in 2.5 % of wins and 2.9 % of draws. What separates them is plain proximity: the further black king is within four squares of the white one in 24 % of won placements and 5 % of drawn.

So the superlinear climb to $n = 14$ was not a growth law but the approach to a transition, sharper than KQKR's: there the drawn fraction went from 0.13 % to 26.6 %, here from nothing to 93 %. Both halves are computed twice, by the symmetry-reduced table and by an unreduced solver sharing no code with it, agreeing at $n = 8$ and $n = 12 \dots 20$.

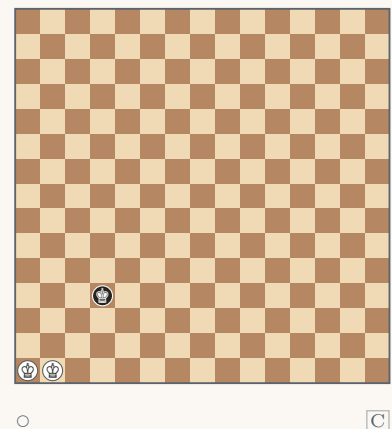
One placement, watched across the transition. The three squares exist on every board from 6×6 up, so the only thing changing is the room behind the black king.



8×8 : **White wins**, 31 plies



14×14 : **White wins**, 109 plies



15×15 : **drawn**

n	6	7	8	9	10	11	12	13	14	15	16	17
<i>Ka1 Kb1 vs kd4</i>	17	25	31	43	49	61	71	89	109	draw	draw	draw
<i>Kf6 Kg6 vs kb1</i>	17	23	25	29	33	35	43	49	55	73	73	73

Two placements, two fates: the first runs away, 17 plies to 109, then stops being a win at all; the second climbs gently and locks at 73 for ever. Not the corner, since both kings end up in one — the second is *already caught* and the first has to be fetched.

Three quarters of the win evaporates

The 14×14 win spends its 109 plies driving the black king from *d4* east along the fourth rank, up the east side and west along the top, where he falls on move 55. Ask the 15×15 table what each position in that drive is worth: **the first 81 plies are drawn**, and the line becomes a win only at ply 82, Black by then pinned in the corner. So the endgame has two phases and only one depends on the board — the finish is local and works at $n = 15, 16, 17$ alike; the drive needs a small board.

Black is not saved by the new ground. Where he is pinned on the top rank of the 14×14 win, white kings *i11* and *j11*, black king *j14*, his five legal squares *all lose*; on 15×15 those same five are *all drawn*, and of the three squares the extra rank gives him, *j15* loses. He is saved because the old ground has stopped being a wall: *j14* is the edge on one board and an ordinary square on the other.

The hunt has a reach, and it is 6. White-to-move placements by the distance from the black king to the nearer white king:

separation	1	2	3	4	5	6	7	8	≥ 9
14×14 , won	267 852	449 696	541 092	552 000	502 780	420 192	333 396	253 872	<i>all</i>
15×15 , won	356 888	16 000	11 192	5 820	2 328	520	0	0	0
16×16 , won	466 380	15 300	10 848	5 776	2 328	520	0	0	0
17×17 , won	599 232	14 796	10 576	5 728	2 328	520	0	0	0

At 14×14 the reach is the whole board, out to 13, with not one drawn placement; at 15×15 it collapses to **six**, and no won placement there or above has the black king seven or more from both white kings. What is left is a fixed set rather than a shrinking share of a growing one: **2328** at separation 5 and **520** at 6 on all three boards, the same to the unit. Reach 6 is necessary and not sufficient (*Ka1 Kb1 vs kd4* has separation 3 and is drawn), but past 6 the question is closed.

Area is irrelevant. The shorter side is everything

$K+K$ vs K solved on rectangular $n_f \times n_r$ boards by a separate program — no symmetry, no index, value iteration to a fixpoint — checked against the tables above on every square board from 5×5 to 14×14 , position, depth and census alike:

board	squares	shorter side	$\lfloor (s-1)/2 \rfloor$	won	deepest
13×13	169	13	6	100 %	99
14×14	196	14	6	100 %	123
15×14	210	14	6	100 %	141
18×14	252	14	6	100 %	197
24×14	336	14	6	100 %	299
20×13	260	13	6	100 %	183
15×15	225	15	7	6.99 %	73
20×15	300	15	7	5.10 %	73

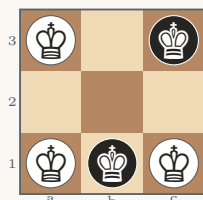
A 24×14 board has 336 squares, half as many again as 15×15 , and **every placement on it is won**, in up to 299 plies. A 15×15 board has 225 and nine in ten of them are drawn. Size does not decide this.

The rule is the shorter side, and the threshold is 14. Every board whose shorter side is at most 14 is won from every placement; every board whose shorter side is 15 collapses to the same local traps and the same frozen 73 plies, 20×15 's deepest survivor being the 15×15 position exactly. The long side may be anything. Read as a distance it answers itself: on a board whose shorter side is s the furthest the black king can get from an edge is $\lfloor (s-1)/2 \rfloor$, which is **6** for $s = 13$ and $s = 14$ and **7** at $s = 15$. Two kings can work on a king at most six from an edge. Whether this 6 is the reach's 6 is not shown.

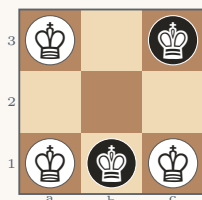
The last column also disposes of an argument used earlier, that steeply climbing depth signals a transition ahead: on 24×14 the depth is 299 plies and still climbing with every file added, with no transition ahead.

9.2.4 Captures decide it

On 3×3 and 4×4 the extra king is worth nothing unless you also have the move. They are the only boards on which Black wins anything, and on both the same placements are a White win with White to play and a Black win with Black to play.



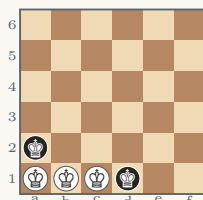
White to move: White wins in 3 plies



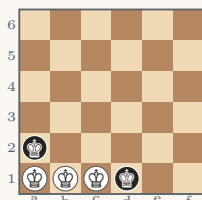
Black to move: BLACK wins in 5 plies

The same five men on the same nine squares, and the move is the whole game. Twelve placements on 3×3 behave this way and **404** on 4×4 , where $Ka1 Kd1 Ka3$ vs $kb1 kd3$ is a win in 9 with the move and a Black win in 17 without, the deepest Black win here. From 5×5 up Black never wins again, and the move is never worth *less* than nothing to either side, so **this endgame contains no zugzwang at all**. §6.2.1 says the same of two knights, for another reason.

Every quiet black move hands White a white-to-move placement, and those are all won, so Black is drawn *precisely* when he has a capture leading to a drawn $K+K$ vs $K+K$. That is the entire defence, and it is sharp: the 6×6 placement below gives him ten legal moves, of which **three** take a white king and only one draws.



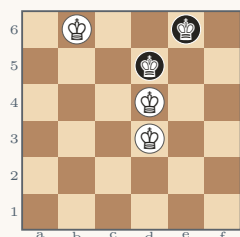
1... $ka2 \times a1$ **draws**: the bare kings stand on a1 and c1, two apart



1... $ka2 \times b1$ **loses in 19**: they stand on b1 and c1, and touch

One square's difference in which king Black takes, and the same trades run off the board into two bare kings that touch or two that do not — and K vs K is drawn *unless* they already stand beside each other. The king it is right to take is the one buried in the corner, which looks least worth having. The share of black-to-move placements this saves falls with the board, 51.2 to 29.8 per cent from 5×5 to 10×10 .

White's trades are the converse. Since any capture wins on the spot an even exchange is not even: it takes 3 vs 2 down to 2 vs 1, which is won. So in the longest win on the board White walks into the trade on purpose.



6 × 6, White to move, wins in 13

The 6 × 6 deepest win — $Ka1 Kf1 Ka6$ vs $kd2 ka4$, 39 plies — caught at ply 27, after thirteen moves of herding. Two plies, no change in material, and the endgame is over: three against two has become two against one, won from every placement there is (§9.2.3). With kings only a trade changes not the *balance* but which endgame you are in.

And White can never manufacture the first capture. Call a placement *quiet* when no white king stands beside a black one. From 5×5 to 12×12 , either side to move, White wins *not one* — 84.2 million of them at 12×12 and the count is zero. So **every white win begins with something already hanging**, and this material is a race for the first capture, decided before either side moves.

That also disposes of the mirror defence: Black *can* answer every white move with its image, but from a quiet position he never has to. The only *symmetric* quiet win is on 4×4 , $Ka1 Kd4$ against $kd1 ka4$, 13 plies for whoever moves; the same idea on 3×3 flips sign, whoever moves *losing* in 4, and on 5×5 it is a dead draw.

Raise the board by one file and it all evaporates. Not because Black keeps out of the way — from a quiet position he usually *cannot* retreat to another — but because he does not need to. Two black kings side by side defend each other, so a white king that takes one is recaptured, leaving K vs K . Linking them is nearly sufficient on its own: with the kings adjacent White’s share over *all* placements runs 21.5 down to 1.0 per cent at $n = 4 \dots 9$, against 34.8% unlinked on 8×8 , and the survivors are all the same jam.

9.2.5 Where it must break

Everything above rests on one clause, *any capture White can make wins on the spot*, and that clause is a statement about $n \leq 14$. It expires.

	13	14	15	16	17
$K+K$ vs K , white to move	all won	all won	7.0% won	6.0%	5.3%
$K+K+K$ vs K , white to move	all won	all won	all won	all won	all won
$K+K+K$ vs K , black to move	all won	all won	8.9% drawn	8.0%	7.2%
$K+K$ vs $K+K$, white to move	15.1% won	13.3% won	0.03% won	0.02%	0.01%

Two kings stop beating one at 15×15 . **Three kings still beat one** — all 415 850 400 white-to-move placements at 15×15 are won, and 16×16 ’s and 17×17 ’s too — but the black-to-move half goes, and every new draw is the same draw: Black takes a white king and reaches the $K+K$ vs K that has just stopped being won. That is precisely the recapture the five-man win assumed could be ignored.

Black’s side of it gets easier at the same moment and by more. His entire defence is a capture into a *drawn* $K+K$ vs $K+K$, and that endgame’s white-to-move drawn share runs 84.9%, 86.7%, **99.97%** at $n = 13, 14, 15$: up to 14×14 he must choose his capture, past the transition he barely has to look. It costs White time in the half he still wins — $K+K+K$ vs K runs 31, 37, 47, 55, 65, 73, 81, 91, 103, 119 for $n = 8 \dots 17$, steps of +6 to +10 up to 15×15 and then +12 and +16. The turn begins exactly where the endgame underneath it broke, as one would expect if these transitions propagate up the lattice.

What is settled here, and what is not

$K+K+K$ vs $K+K$ is a White win from every white-to-move placement for $3 \leq n \leq 11$ — a computation, not an argument, and it does not extend past 14×14 by the reasoning above, because the reasoning stops being true there. Whether the endgame survives its own transition is **not settled**, and the depth data does not settle it either way: by §9.2.1’s measure the five-man line is flat at $\alpha \approx 2.05$ and shows none of the last-window jump both collapsing endgames made, but the clause the win rests on is false at 15×15 whatever the depth is doing at 11×11 . What would settle it is out of reach by four board sizes — the 15×15 table is 5.9×10^9 canonical slots per side to move, 24 GB at two bytes an entry, against the 2.4 GB that 12×12 costs.

10 Capture-the-king chess, and the other rule sets

In chess the objective is to *capture* the king, and mate is the name for a capture that cannot be prevented. Taking that literally is the second rule set every section above is measured under.

The rules: every king is a man, and you win by taking them all

Let each side have $k \geq 1$ kings. Everything else is chess.

1. Every man moves and captures exactly as in chess, kings included: one square in any direction, taking whatever stands on it.
2. **A king is an ordinary piece in respect of check.** It may move to an attacked square, it may be left standing on one, and it may be captured like any other man. There is no check, no checkmate and no pin: *every* move that obeys rule 1 may be played.
3. A player wins the instant the opponent's *last* king is taken. The game ends with that capture and nothing that would have followed it matters — in particular a player may take the last king with a move that leaves their own king hanging. Losing a king while others remain costs a piece and nothing more.
4. A player who still has a king and no legal move is stalemated and the game is drawn. Under rule 2 that requires *total immobility*, which §10.1.1 characterises; for the material in this document it cannot happen at all.
5. Promotion and *en passant* are as in chess, promotion being to queen, rook, bishop or knight. Castling plays no part.

Nothing here privileges one king over another, or Black's kings over White's.

10.1 One king a side: chess, up to stalemate

Rule 2 sounds like a large change and rule 3 like a small one; with one king a side it is the other way round. Call a position **sound** when the side *not* to move does not have its king attacked — the condition every chess-reachable position satisfies, and the one every table here uses to decide which slots are legal.

Theorem. With one king a side, capture rules are chess with one clause changed

Let P be a sound position with one king a side, and let $\text{cap}(P)$ and $\text{chess}(P)$ be its values under the rules above and under the ordinary rules. Then $\text{cap}(P) = \text{chess}^*(P)$, where chess^* is ordinary chess with exactly one clause replaced: *when the side to move has no legal chess move, the game is drawn if that side has no move under rule 2 either, and otherwise that side loses.*

So the two games agree except where the side to move has no chess move, and there they differ in two ways and only two: **stalemate becomes a loss**, the familiar case; and **a checkmate becomes a draw**, when the mated side has no move under rule 2 at all. Ordinary checkmate is otherwise still a loss, and ordinary stalemate with the mover totally immobile is still a draw.

Proof. (i) In a sound position the side to move does not attack the enemy king, so it has no king capture available; a king capture can only follow a move that left the mover's own king attacked. Call such a move *self-exposing*. The moves that are not self-exposing are exactly chess's legal moves, and they lead to sound positions again. (ii) A self-exposing move loses at once: some enemy man can move onto the king's square, rule 2 makes that legal, and rule 3 ends the game there. (iii) So if the mover has a chess move, deleting the self-exposing moves changes nothing and what remains is the chess move list; if it has none but has some move, every move loses by (ii); if it has no move at all, rule 4 draws. Backward induction over the sound positions — a finite graph, once the same repetition rule is given to both games — yields $\text{cap} = \text{chess}^*$. $\square$

Hence the corollary that makes the two games feel alike: **optimal play never gives a king away from a position that is not already lost.**

10.1.1 Which stalemates survive

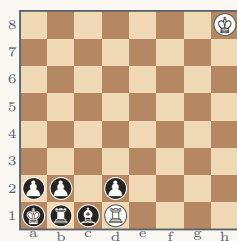
Corollary. The surviving stalemates are exactly the totally immobile positions

The side to move is stalemated exactly when *no man of that side has any move at all*: (i) every non-pawn man, its king included, has every square it could reach either off the board or occupied by a man of *its own* colour — an enemy man there would be capturable, hence not a blocker — and (ii) every pawn has the square in front occupied and no enemy man on either capture square.

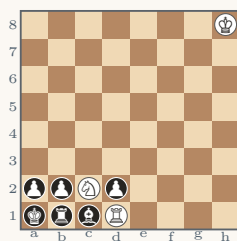
Two consequences. **The immobile side needs at least four men**, since by (i) every square beside its king holds one of its own and a king has at least three of those. And **if the immobile king is attacked, it is attacked by a knight**: everything beside it is occupied, so a line piece would have to bear through an occupied square and a pawn or king would have to stand on one.

So they fall into two families, the second being the whole of the theorem's second bullet: **(S)** the immobile king unattacked, which both rule sets draw; **(M)** the immobile king attacked, necessarily by a knight, which ordinary chess calls *checkmate* and these rules call a draw. It is the only case where capture rules are kinder to the defender.

Both families are inhabited; the diagrams differ by one white knight.



(S) black to move, unattacked and with no move at all: a draw under both rule sets



(M) the knight checks a1: *mate* in ordinary chess, stalemate and a draw here

Black is immobile in both, man by man: the king shut in by its own rook and pawns; the rook on b1 with a1, b2, c1 all its own; the bishop on the rim with only b2 and d2; the pawns blocked, with nothing but their own men to take, and d2 blocked by a rook a pawn cannot capture. In (M) the knight on c2 is untouchable too, and gives the check no line piece could give.

None of this can happen to the material in this document

Immobility needs four men on the immobile side. Every side of every table here has two: king and queen, or king and king. So rule 4 never fires, and every table sees only the first half of the theorem — the half that turns the stalemate draws of ordinary chess into wins. That is exactly why **K + Q vs K** is won here in 17 plies against 19 in the ordinary game.

Checked against the ordinary table. `./egtb kqkk --capture --theorem` walks every K + Q vs K placement for $3 \leq n \leq 9$ against ordinary KQK, which this program builds with a different solver. Every chess win stays a win, every loss a loss, and **every chess checkmate and stalemate has capture value exactly -2**: on 8×8 , 364 checkmates and 872 stalemates, with 22 176 chess draws staying drawn — the published KQK census to the man, so three components agree. The last count says the drift does not propagate.

10.2 More than one king: the mating rules

§7 armed the defender; this does the opposite in the same slot. Black's second man is a second *king*: it can never attack anything, the two may stand beside each other and each blocks the queen's rays for the other, and neither may stand on or beside the white king.

A mate counts only when both kings are mated at once

That sentence is the endgame, and it has two readings. They differ in one predicate, so the program computes both.

Strict. After Black's move neither king may be left attacked — the ordinary rule of chess, applied to each king in turn. Mate is both kings attacked with no move of either escaping. *No move at all, with only one king attacked, is stalemate* and a draw: what the ordinary rule would call mate, this one declines to count.

Loose. A black king may stand in check. What Black may not do is leave *both* kings attacked at once.

Everything else survives, which is why this endgame keeps the cheap machinery of §1 while KQKR does not; only the *index* changes, the block key becoming a king *triple*. A third reading drops mate altogether and plays for the kings themselves: the capture rule above.

11 How far any of this can be trusted

Bellman re-derivation	Every entry recomputed from its successors with an independent forward move generator: $n \leq 24$ for KQK and KRK, ≤ 14 for KBBK and KBNK, ≤ 16 for KQKR, ≤ 17 for KQKK, ≤ 11 for KQKK under capture rules, ≤ 16 for KNK and KNNK, ≤ 10 for KNNNK, ≤ 8 for the kings-only lattice — there all six tables at once, a five-man entry being only as good as the entries it reads. Larger boards on a sample.
Unreduced reference solver	The same endgame solved with <i>no</i> symmetry, by a different algorithm, compared position by position: 33 030 144 placements at $n = 8$ for each bare-king endgame; every n^8 ordered placement to $n = 6$ for KQKK and its four capture tables, which checks the king-triple index at placements reachable only through a symmetry; every ordered placement to $n = 7$ for KNNNK, 530 832 288 of them; and for §9 the whole lattice re-solved as a single hash of every state, testing ranking, D_4 reduction and conversion seeding at once. This is what catches indexing bugs, which a self-consistency check structurally cannot. Both bugs the capture rules cost were in the <i>checker</i> : a solver that sweeps until nothing changes stops too early on a table that converts, and one that takes a win as soon as it is computable records the first win rather than the shortest.
Published 8×8 figures	462 canonical king pairs, 364 KQK checkmates, 872 stalemates, 22 176 queen- <i>en-prise</i> draws and KQKR's mate in 35, all reproduced exactly; KRK, KBBK and KBNK then <i>derived</i> rather than looked up. KQKK, KNNNK and the kings-only tables have no published figure to check against. KNNNK has instead the unreduced solver over all 8^{10} ordered placements at $n \leq 7$, and the 120 KNNK mates were counted independently by a throwaway script.
Naive generator and index	For KQKR and KQKK, where brute force runs out of memory long before the interesting boards. Each move generator is compared against one that walks every ray a square at a time; KQKR's index against the definition of a D_4 orbit, KQKK's against the requirement that a value be constant on a symmetry class, exchange of the two black kings included. Random placements to $n = 16$ and $n = 12$.
A second, independent solver	$K+K$ vs K is a table of §9 and also K vs $K+K$, which §9.2.3 builds out of the four-man capture solver: different index, generator and induction. The two agree on the win count, the deepest win <i>and</i> the white-to-move draw count from 8×8 to 20×20 — on the climb, on the collapse, and on the 73 plies frozen after it. Two solvers sharing no code agreeing on where a transition sits is the strongest evidence here that it is real. A <i>third</i> , <code>tests/rect_kk.cpp</code> , reproduces the census on every square board to 14×14 before being pointed at the rectangles of §9.2.3.
A guess not made	KNNNK's sub-table could have been assumed instead of built, and the assumption would have been wrong. The solver refuses to run without a KNNK table attached.

The bug that mattered

KBNK first reported mate in **35** on 8×8 . The textbook value is 33, and that discrepancy is what exposed it.

Phase B treated “the black king steps onto a white piece” as a capture, hence a draw. With one white piece that is sound; with two it is wrong. And it *hid*: two bishops can only defend each other when same-coloured, and those positions are drawn anyway, so the bug was live in KBBK for as long as KBBK existed and changed not one entry. A bishop and knight defend each other freely, and there it cost 2816 lost positions on 5×5 . The verifier and the unreduced solver flagged the same positions independently — the wrong answer being a plausible number rather than a crash is the argument for keeping both.

All figures computed by the generator in this repository on an Apple M2, 8 cores, 16 GB.

Depth data: `artifacts/results/*-max-dtm.csv` · census: `artifacts/stats/`

KQKR: `src/kqkr.cpp` · KQKK: `src/kqkk.cpp`, `src/indexkk.hpp` · capture rules: `src/kqkkcap.cpp`

Kings only: `src/kings.cpp` · rectangular boards: `tests/rect_kk.cpp` · the mating rule: `src/policy.cpp`

Same-coloured bishops: `kqk bishops --capture --stale`